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Design of an Automated Silicone Rubber Dispenser

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Figure 1: Rendered CAD model of Design

Abstract

This report presents the design of an automated silicone rubber dispenser developed for soft robotics research at KU Leuven. The dispenser addresses the limitations of manual silicone mixing, which is time-consuming, prone to inaccuracy, and difficult to reproduce. The machine is designed to dispense up to four individual silicone components with a target accuracy of ± 0.1 g per component, within a time window of 2 minutes 30 seconds per component, for total dispensed weights between 5 g and 100 g.

A structured design process was followed, starting from a functional decomposition of five core machine functions — storage, circulation, actuation, amount control, and user interface — followed by concept generation and a weighted value analysis. Three concepts were evaluated: a gravity-fed system, a syringe-actuated system, and a peristaltic pump system. The peristaltic pump concept was selected based on its superior scores in dispensing accuracy, user-friendliness, low maintenance, and built-in circulation capability.

The final design consists of four custom-built peristaltic pumps driven by NEMA 23 stepper motors, paired with servo-actuated stopcock valves that switch between dispensing and circulation modes. Silicone is stored in 3D-printed containers positioned above the pumps to exploit gravity-assisted flow. A 200 g load cell with HX711 amplifier provides weight feedback to a Raspberry Pi 4, which runs a touchscreen interface and an iterative dispensing control algorithm. The total component cost amounts to approximately €856.

Testing revealed that the load cell introduces a systematic calibration offset and that minimum droplet size limits dispensing precision. The measured process capability index C_p of the complete system is approximately 0.26, falling short of the ± 0.1 g specification. The primary contributors to error are load cell accuracy and residual flow after pump stoppage. Recommendations for further development include improved load cell calibration, chemically compatible and optimised tubing, and refined valve sizing.

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1 Introduction and Requirements

The first part of this chapter frames the problem in the current context around Soft Robotics. The second part gives a more detailed summary of the technical requirements and goals of the project.

1.1 Introduction

Soft robotics is a branch of Robotics that aims to make robots less harmful/dangerous by using soft actuators. These are made by mixing multi-part curing silicones (e.g. Part A and B silicone). After being poured into a mould, the different parts react and cure to form the desired component. With these soft components, soft machines can be made. An example can be seen in Figure 2.

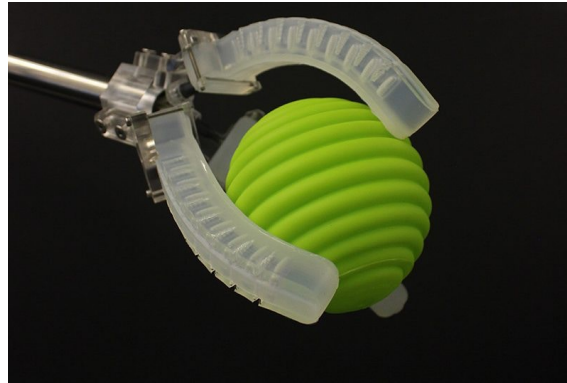


Figure 2: Example of a Soft Robotic Gripper [1]

Currently when mixing two-part silicone rubbers, the two parts are poured by hand with a precision scale from a big bucket (see Figure 3). Furthermore, to get the desired hardness, multiple two-part silicones are combined. This means that the relative weights of each of the (up to four) parts contribute to the properties of the final silicone. Needless to say, this means that precise mixing of these components is crucial to guarantee the repeatability of various experiments using these silicones. Small variations in the composition can already cause differences in material behaviour.



Figure 3: Example of Silicone Pouring by Hand [2]

It is easy to see from Figure 3 that accurate weighing of the different parts of the silicone is not easy to do by hand. Since the silicone also begins to cure as soon as the two parts touch, there is a limited amount of working time available for further processing (such as degassing) before it becomes too viscous to work with. Additionally after long non-usage, the individual parts of the silicone need to be mixed to negate separation of heavier molecules and/or keep it from slightly hardening. Lastly spillage/contamination is a problem when measuring by hand.

This process clearly isn't optimal. It is time consuming, accurate to a certain degree, but not easy or reproducible. Therefore the project goal is to design an easy to use machine (with software) that can take a user input (e.g. amount of silicone of a certain hardness), and output the desired silicone into a cup with high accuracy in a reasonable time frame.

Chapter 2 will discuss the generated ideas and design methods. At the end of Chapter 2, three main concepts are compared and a final concept is chosen. This final concept is then detailed in Chapter 3, where each component is described and an explanation is given about how the machine works. The last chapter discusses the obtained results and the future of the project. Upgrades that can be made and possible problems that can still be solved.

1.2 Requirements

In this section, a short overview of the most important parameters (in bold) to optimize is given. These are a combination of conditions specified by the client with some boundary conditions.

1.2.1 Performance

- **Dispense weight:** A total weight between 5g-100g should be possible to dispense. The weight of the individual components is therefore 50g maximum. As an example: 100g of silicone = 50g Part A + 50g Part B. The process is used most in the 30g-40g range.
- **Accuracy:** Each component of the final silicone (up to four parts) cannot deviate more than 0.1g from it's desired weight. As an example: 50g of A and B required to have 100g of silicone, means that the allowed limit is 40.9g of A and 50.1g of B or vice versa.
- **Circulating:** Each component should be able to be circulated to avoid partial hardening and keep it homogenous before dispensing.
- **Dispensing time:** The dispensing time should be minimized to allow further processing time to degas and properly combine the different components together. The maximum allowed time to dispense each component is 2min 30s. (Based on a working time of 15min - 5min for degassing = 10min and an edge case of four components that need to be dispensed → 2.5min per component)
- **Waste:** The total waste of silicone per pour should be max 5g.

1.2.2 Interface

- **Ease of use:** The machine should be self-explanatory and no extra knowledge should be required.
- The use of a display for clear visualization is preferred.
- Possibility to track the approximate amount of silicone left.

1.2.3 Maintenance

- **Cleaning:** No cleaning is preferred, no mess.
- **Spillage:** No contamination with environment during maintenance.
- **Repairing:** Maximal use of standard commercially available parts where possible. The parts should also be easily accessible to replace.
- **Refilling:** Easily accessible or swappable cartridges to refill silicone.

1.2.4 Additional

- **Budget:** ~ 1000 euro target price, not including some development budget for testing.
- **Dimensions:** Tabletop sized. Ca 75 x 75 x 75 cm³
- Materials in contact with silicone should not inhibit hardening/functionality of the silicone.

2 Design Selection Process

A number of structured methods are employed to address the problem in a systematic manner.

First, a functional decomposition is applied. This method breaks the overall problem down into a set of essential functions that the final solution must fulfil. For each identified function, multiple potential solution principles are generated through a structured brainstorming process. Subsequently, these functions are combined into integrated solution concepts, representing feasible machine designs.

Following this, all concepts are subjected to a value analysis. This approach enables a more quantitative evaluation of each concept based on predefined criteria, allowing for a more objective comparison of their performance and feasibility.

Based on these scores, the three most promising concepts are selected and briefly discussed further. Finally, the concept with the highest overall value is selected for further development and design in Chapter 3.

2.1 Functional Decomposition

The problem can be divided into five essential functions that the machine should have. In this paragraph, they are discussed in more detail and for each function multiple solutions are briefly discussed. A structured overview is given in Figure 4.

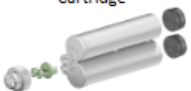
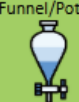
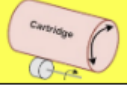
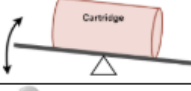
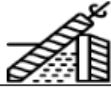
Solutions→ Functions↓	1	2	3	4	5	6
Container	Bag 	Syringe 	Cartridge 	Deformable Tube 	Funnel/Pot 	Bottle 
Circulation	Rolling Cartridge 	External Mixing Machine 	Seesaw Mechanism 	Peristaltic Pump 	Manual 	Stirring Rod 
Actuation	Pneumatic Piston 	Gravity Flow 	Rolling 		Archimedes Screw 	Lead screw 
Amount Control	Flow Sensor (Feedback) 	Weight Sensor (Feedback) 	Model Based (Feedforward) 			
User Interface	Analog buttons + Screen 	Touchscreen 				
	Gravity fed		Syringe		Peristaltic pump	

Figure 4: Functional Decomposition: The different rows represent the different functions that need to be fulfilled, the columns represent possible solutions. The colours indicate the combinations of the three most promising concepts that will be discussed in Section 2.2

2.1.1 Container

The container is the place where the silicone are stored when the machine is not in use. This is also the place where they should be refilled when the machine is empty (and maybe circulated). Possible solutions are:

- **Deformable Bag:** The silicone is stored in a deformable bag with a connection for a tube (e.g. blood bag).
- **Syringe:** The silicone is stored in a medical syringe.
- **Cartridge:** The silicone is stored in a special cartridge, made for multi component silicone such as the MIXPAC F-system cartridge [3].
- **Deformable Tube:** The silicone is stored in a deformable tube, such as a toothpaste tube.
- **Funnel/Pot:** The silicone is stored in a special funnel or pot with connections for tubes.
- **Bottle:** The silicone is stored in the bottle in which the silicone is sold and delivered.

Some illustrations of the less straightforward solutions are given in Figure 5.

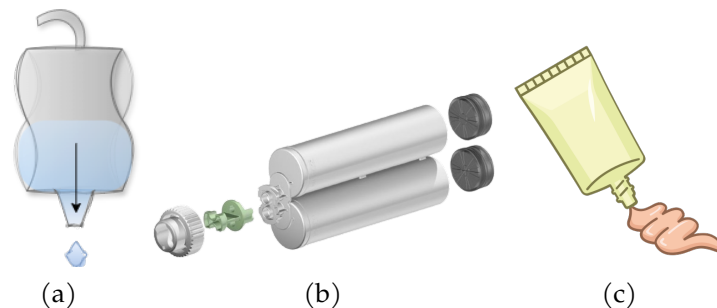


Figure 5: Deformable Bag with Silicone (a), Special MIXPAC F-system Cartridge [3] (b), Deformable Tube with Silicone (c) [4]

2.1.2 Circulating

The circulation is the mechanism that should prevent the settling of the heavier molecules of silicone and the partial hardening after prolonged non-use. Possible solutions are:

- **Rolling Cartridge:** The cartridge with silicone gets rolled via a small wheel to be circulated.
- **External circulating Machine:** The cartridge with silicone must be taken out and placed in an external machine to be circulated.
- **Seesaw Mechanism:** The cartridge with silicone is placed on a seesaw mechanism that gently sloshes it back and forth.
- **Peristaltic Pump:** A peristaltic pump is used to pump the silicone from and to the container in which it is stored.
- **Manual:** The silicone must be stirred by hand to circulate it and automation is omitted.
- **Stirring Rod:** A stirring rod/bar is installed to circulate the silicone in the container in which it is stored.

Some illustrations of the less straightforward solutions are given in Figure 6.

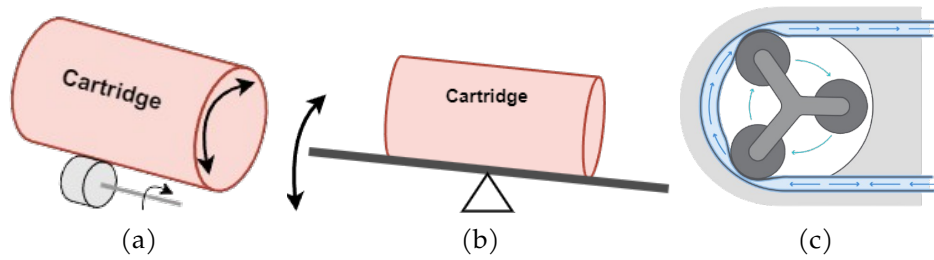


Figure 6: Rolling of the Cartridge with Silicone via a Small Wheel (a), Sloshing of the Cartridge with Silicone with a Seesaw Mechanism (b), Circulation of the Silicone with a Peristaltic Pump [5] (c)

2.1.3 Actuation

The actuation is the mechanism that moves the silicone from the container to the final output place for the user. Possible solutions are:

- **Pneumatic Piston:** A pneumatic piston is used to push against e.g. the back of a cartridge to push the silicone out.
- **Gravity Flow:** A valve is opened from a container and the silicone will just flow out via gravitational forces.
- **Rolling:** Two rolls are used to squeeze the silicone out of e.g. the deformable tube or bag.
- **Peristaltic Pump:** The peristaltic pump can also be used to pump the silicone from the container to the outlet.
- **Archimedean Screw:** An Archimedean screw is used to push out the silicone, similar to a screw extruder for molten polymers.
- **Lead Screw/Rack and Gear:** A lead screw or rack and gear system is used to push the silicone out, similar to the pneumatic piston.

Some illustrations of the less straightforward solutions are given in Figure 7.

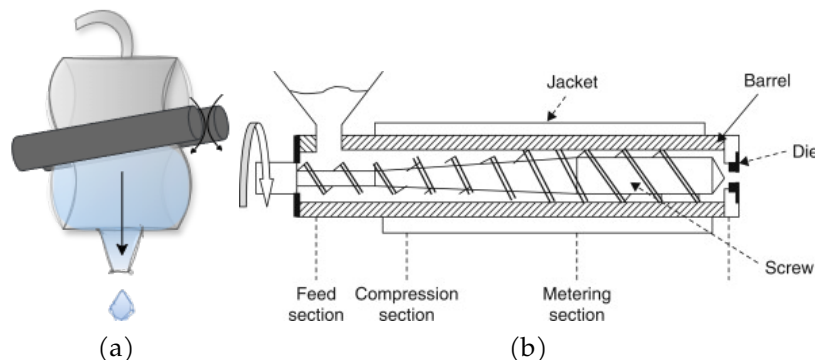


Figure 7: Deformable Bag/Tube between Two Rollers to Squeeze out the Silicone (a), Screw Extruder Similar to the Extruders for Molten Polymers [6](b)

2.1.4 Amount Control

The amount control is the method to measure and control if the correct weight/volume of silicone is dispensed. Possible Solutions are:

- **Flow Sensor:** A flow sensor is installed in-line with the silicone flow to perform feedback and control the dispensed amount.
- **Weight Sensor:** A weight sensor is placed underneath the final dispensing area to measure the dispensed weight and perform feedback and control the dispensed amount.
- **Model Based:** An model of the system is made and used to perform feed-forward control and an extra sensor is omitted.

2.1.5 User Interface

The user interface represents the way the user interacts with the machine and gets information about it. Two (similar) solutions are:

- **Analogue Buttons with a Screen:** A simple LCD-screen is used in combinations with some simple buttons, such as arrows, to interact with the machine and the software.
- **Touchscreen:** A touchscreen is used instead of the analogue buttons to interact with the machine and the software.

2.2 Concepts

In this paragraph, the most promising concepts are discussed. This chapter gives a brief explanation of the composition and working of each concept. The strength and weaknesses will be discussed in the chapter about the value analysis. Each of the following concepts works with a control feedback loop that gets feedback from a weight sensor under the cup.

2.2.1 Gravity Fed

The first concept is the Gravity Fed machine (blue path in Figure 4). The user should screw the bottle with silicone on the threaded holder at the top of the system. After opening the valve, fluid will flow due to gravitational pull. The valve closes when the right amount is reached. The valve is automatically controlled. No internal circulation mechanism is provided, so the user should make sure the silicone is as homogenous as desired. This can be done by putting the bottle in external mixing apparatus. The concept can be seen in Figure 8.

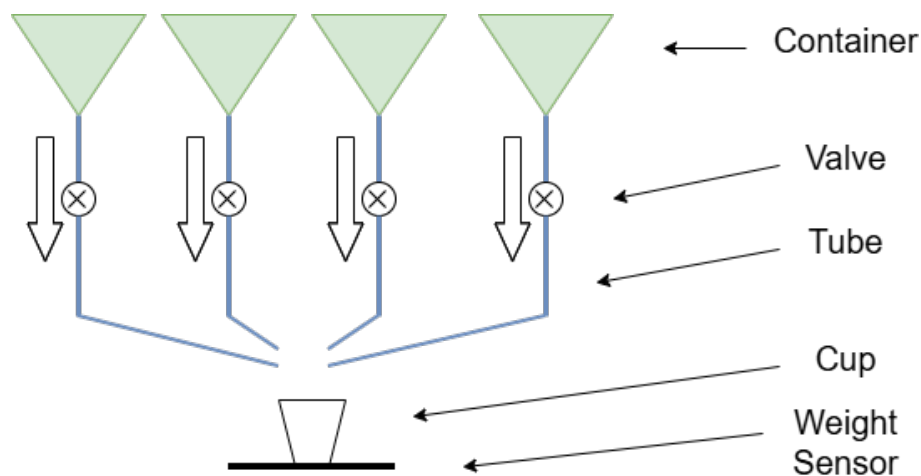


Figure 8: Illustration of the Gravity Fed Concept, the Containers, the Valves, the Tubes, the Cups, the Weight Unit and the Direction of the Flow of the silicone are Indicated with an Arrows

2.2.2 Syringe

The second concept is a machine with syringes as containers (yellow path in Figure 4). The user should first fill the syringes with silicone. The actuators will start to move after entering the desired amount. The actuation is done by a lead screw that pushes the handle of the syringe. The controller automatically stops the actuation after the right amount is achieved. The circulation is done by two rollers that support the syringe. The concept can be seen in Figure 9.

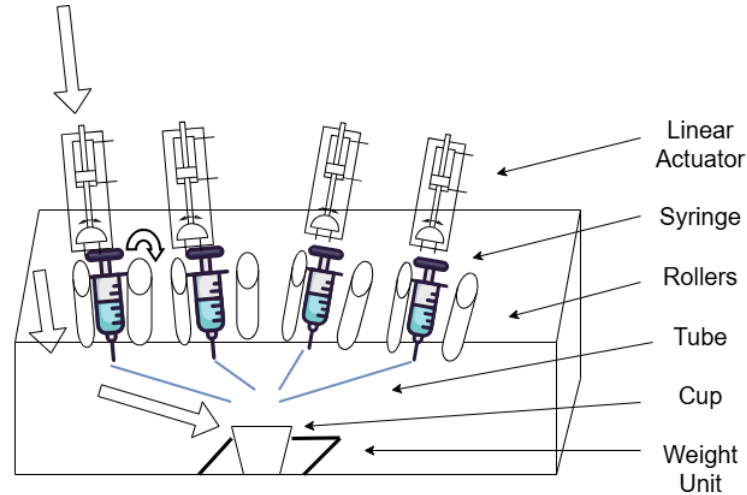


Figure 9: Illustration of the Syringe Actuated concept, the Linear Actuators, the Syringes, the tubes, the Cups , the Weight Units and the Direction of Flow are Indicated with Arrows [4]

2.2.3 Peristaltic Pump

The last concept is a machine actuated by a peristaltic pump (green path in Figure 4). The user should first fill the container with silicone (when the container is empty). The pump will start pumping after entering the desired amount. The controller automatically stops the pumping when the right amount is achieved. The circulation is built in by redirecting the fluid with a circulation unit. The fluid is then fed back into the container. The concept can be seen in Figure 10.

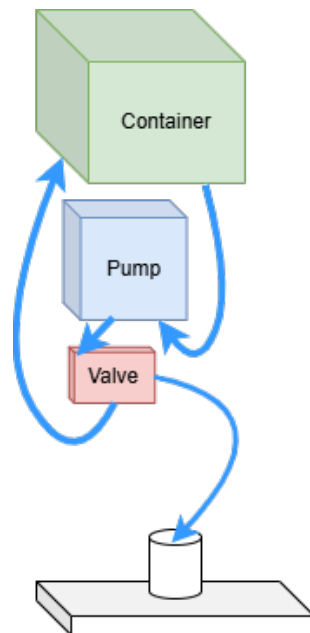


Figure 10: Illustration of the Concept with the Peristaltic Pump, the Container (in Green) , the Pump (in Blue), the Valve(in Red), the Cup and Weight unit(in White), the Blue Arrows Indicate the Flow of Silicone

2.3 Value Analysis

The final decision is made with the help of a value analysis. In this paragraph, the method is first discussed. After that, the line of thought behind the categories is explained. Lastly, the scores of the concepts of the previous paragraph get explained in more detail and the conclusion will be elaborated.

2.3.1 Method

The value analysis is a method to quantitatively analyse the obtained concepts, to decide which one is best. There are three main categories on which each concept gets rated. The categories are subdivided into smaller criteria. Each criteria gets assigned a weight. Five is the highest weight (most important), one is the lowest weight (less important). Every concept gets a score on each criteria on five. Afterwards, a weighted sum gets calculated, and the final score decides which concept is the best.

The scores are not given absolute. They are given relative to reference concepts. These are two concepts which are currently in use in the lab of the KU Leuven. A survey was sent to the people who use them to rate their accuracy, precision and pouring time to have a more concrete comparison.

Table 1: Table with Value Analysis, Concepts on the Left, Categories on the Top, Weights beneath Categories, Scores on the Right

Categories	Mechanical working	User friendliness		Other			Score
	Accuracy	Involvement	Maintenance	CAPEX	OPEX	Silicone Waste	
Weight	5	5	1	1	1	3	0
Reference 1: Manual pouring	5	0	5	5	5	3	49
Reference 2: Dispensing gun	0*	2	4	4	4	0	22
Concept 4: Cartridge system intern mixing feedback	3	4	3	2	3	3	52
Concept 8: Syringe system intern mixing feedback	4	3	3	2	3	3	52
Concept 9: Peristaltic pump feedforward	2	4	3	4	4	4	53
Concept 10: Peristaltic pump feedback	4	4	3	3	4	4	62
Concept 11: Gravity fed feedback	2	4	4	4	4	4	54

2.3.2 Categories

- Mechanical working: The mechanical working contains Accuracy. Because accuracy is one of the most important requirements, it gets a weight of five.
- User friendliness: User friendliness contains involvement and Maintenance. Because the goal was to make the task automatic, involvement has a high score. Maintenance was an aspect to take into consideration, but was not the most important one.
- Other: The remaining criteria are CAPEX, OPEX and silicone waste. Both CAPEX (Capital Expenses) and OPEX (operational expenses) are less important relative to the others. A budget was given, but no other specifications are demanded about price. Silicone waste is somewhere in the middle of importance, because it is more important to minimize the waste than to have a low maintenance, cheap machine. But it is less important to trade accuracy and involvement for less waste.

2.3.3 Concepts

- Gravity fed system with feedback:

The main advantage of this concept is that it is very simple in comparison to the others. It is easy to understand, maintain and repair, which where important requirements that where discussed before. The simplicity also makes that the machine is cheap.

The disadvantages are that the control is perhaps too simple, which reduces the accuracy. After a test, it was clear that the flow rate of this setup was too low to fulfil the required time window.

- Syringe system with feedback:

The main advantage of this system is its accuracy. The lead screw actuation, in combination with a stepper motor with micro stepping, can produce very small increments in displacement of the fluid. This increases the accuracy of the system.

The drawbacks are that the syringes have to be refilled quicker then the other concepts. To generate the same pressure, a bigger syringe usually requires more force. The involvement scores lower, because the frequent refilling. This also increases the operational cost, because after use, the syringes can not be recycled. The silicone waste also scores high because when refilling the syringe, things could get messy. The CAPEX is also higher, because there need to be a linear actuator for each syringe, and actuators to drive the rollers for the mixing. This makes it more expensive.

- Peristaltic pump system with feedback:

The advantages of this system are its involvement and maintenance. The use of a large pot on the top, reduces the frequency to refill. The long lifetime of the tubing, makes that the machine can go long without maintenance. Due to clever design of the pump, and the use of Luer-locks, the maintenance will be intuitively and easy. The accuracy is increased by reducing the tube diameter near the outlet of the machine. The possibility to easily reverse and stop the fluid movement by reversing the pumping direction is also a major positive aspect for accuracy. By emptying the tubing and pot completely, and the option to reverse the flow easily, the spillage of silicone is also minimized.

The disadvantages are smaller relative to the advantages. There are still two actuators for line (one to control the pump , one to control the valve) , which makes the initial investment higher. This amount of actuators also increase the power consumption of the machine, so the operational cost increases.

2.3.4 Conclusion

As seen in Table 1, the peristaltic pump concept gets the highest weighted score. It is clear that the benefits of this concept outweigh the drawbacks. In the next chapter, the design of this concept will be discussed in more detail.

3 Design

In this section the full, detailed design of the 'Peristaltic pump'-Concept from Section 2.2 is explained and illustrated. First, a general overview is given to illustrate the dispenser and workflow as a whole. The different parts of the design are then discussed separately and related to the requirements.

3.1 Working principle

The general principle is the same as explained in Section 2.2.3, but is further explained here based on a model sketch in Figure 11 of the complete system. The final rendered system can be seen in Figure 1.

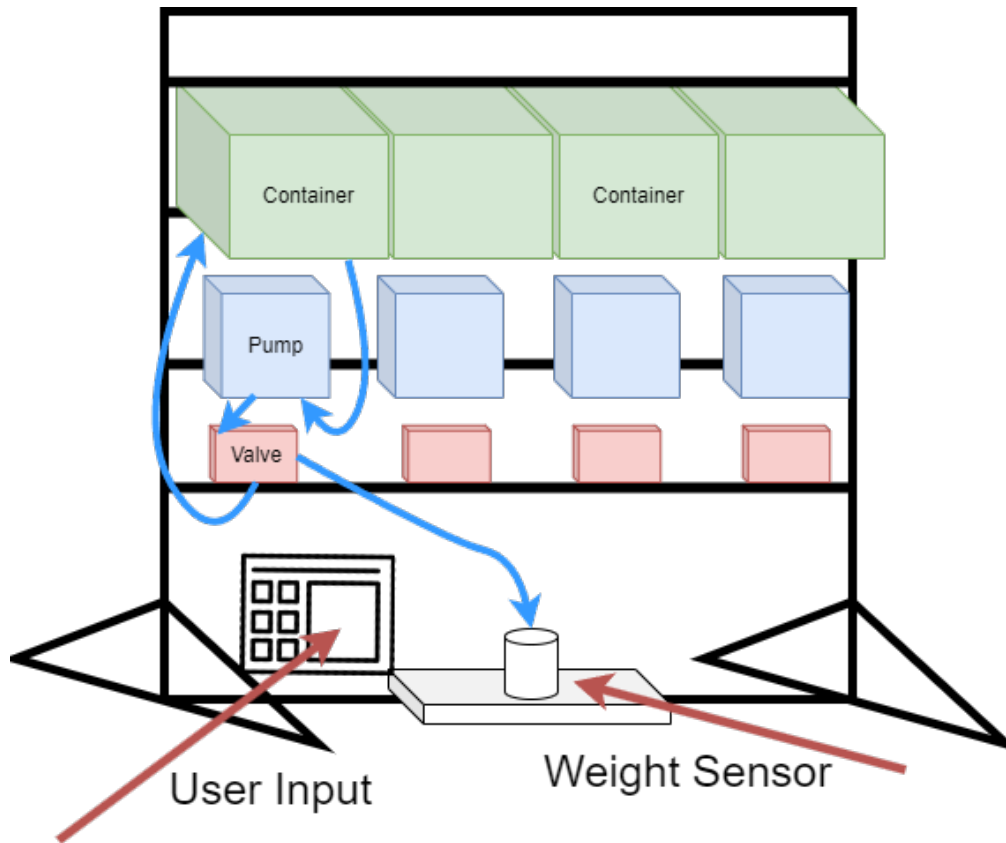


Figure 11: Model Sketch of the Design of the Silicone Dispenser to Illustrate the General Working Principle: Four Containers with the Individual Parts of Silicone (in Green) are Connected to four Peristaltic Pumps (in Blue) that Pump the Silicone via a Valve (red) to a Weight Sensor. The Blue Arrows Indicate the Flow of Silicone.

The successive steps to dispense silicone with the machine are as follows:

1. The user gives an input to the machine via the software/interface on a screen with the help of a rotary encoder. The user can select different modes, the two most important ones are: circulating, to circulate the silicone on a regular basis and keep it from partially hardening, and dispensing, to dispense a precise amount of silicone.
2. The software sends a signal to the pumps and valves based on the input of the user. This actuates the pumps and can switch the valves to the desired position based on the selected modes.
3. Finally the pumps are activated and begin pumping the silicone. Each pump is connected to a container, in which a single part of a certain silicone is stored. As explained in Section 1 up to four individual parts may be required to dispense a certain hardness of silicone, therefore four pump-container-valve combinations are installed.

4. While dispensing each part of silicone, the weight sensor measures the current weight and is used to give feedback to the pump and regulate the dispensed amounts.
5. When the desired amounts of silicone are reached, the machine stops and indicates on the screen that it has finished dispensing.

As an example if 'dispensing of 20g of silicone with a certain hardness' is entered, the precise amounts of the (up to four) individual parts of the silicone is calculated. A signal is given to the valves to switch to the dispensing position connecting the container via the pump to the final dispensing plate with the weight sensor. Finally the pumps are actuated individually to dispense the desired weights of their corresponding parts of the silicones.

In the following sections, each (major) component of the design is further explained in detail.

3.2 Design Pumping System

This section discusses the design of the peristaltic pump, the tubing, container and connections of the chosen concept from Section 2.2. The basic principle of a peristaltic pump is to push a fluid via the progressive deformation of a flexible tube. This is typically done by placing a flexible tube in a semi-circular casing that gets compressed by a couple of rollers attached to a motor. As the rollers move and compress the tube, the fluid is pushed forward and a flow is generated [5, 7]. An illustration of the basic principle and shape of a peristaltic pump is shown in Figure 12

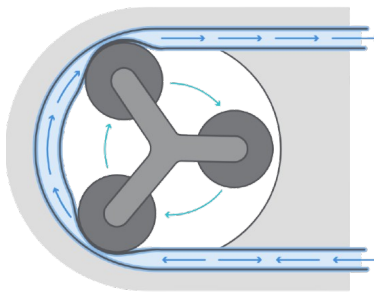


Figure 12: Illustration of the Principle of a Peristaltic Pump: a Couple of Rollers Compress a Flexible Tube in a Housing to Push a Fluid Forward [5]

The main advantage of a peristaltic pump is that the fluid never comes into direct contact with the pump itself, only the flexible tube. A consequence of this system is that the tube can be replaced without cleaning the pump itself. Furthermore, peristaltic pumps can handle viscous liquids and can be very accurate [7, 5, 8]. These properties comply perfectly with the requirements listed in Section 1.2, primarily: pumping viscous silicone, ease of use, no cleaning and dispensing accuracy.

Due to the large amount of parameters influencing the working of a peristaltic pump, such as the amount of rollers, tube diameter, fluid viscosity, occlusion... [9, 10], a custom design was developed. This allowed for full control over all design parameters, without being constrained by the limitations of a commercial solution that would be difficult to modify. This was done mostly via additive manufacturing, primarily using PLA and PETG as they have reasonable mechanical properties. Furthermore the designs were made to minimize supports. This means it is also easily replaceable since the parts can be reprinted on most (cheap) commercial 3D-printers.

The most important requirements and boundary conditions for the design of the pumping system are repeated here:

- **Dispensing Time/Weight:** maximum 2min30s per component (for max 25g/component)
- **Accuracy:** an accuracy of $\pm 0.1g$ per component
- **Silicone Viscosity:** Commonly used two-part silicone has a viscosity of up to 13000cP [11] (viscosity of water $\approx 1cP$)

The full design of the pump is shown in Figure 13. It is based on a Nema 23 hybrid stepper motor [12]. Stepper motors are particularly well-suited for this application as they allow precise, incremental angular displacement, which directly translates to accurate and repeatable fluid dispensing. Furthermore, the NEMA 23 form factor provides a substantial torque margin, ensuring that the motor would not become a limiting factor when testing different tubing configurations or rotor geometries.

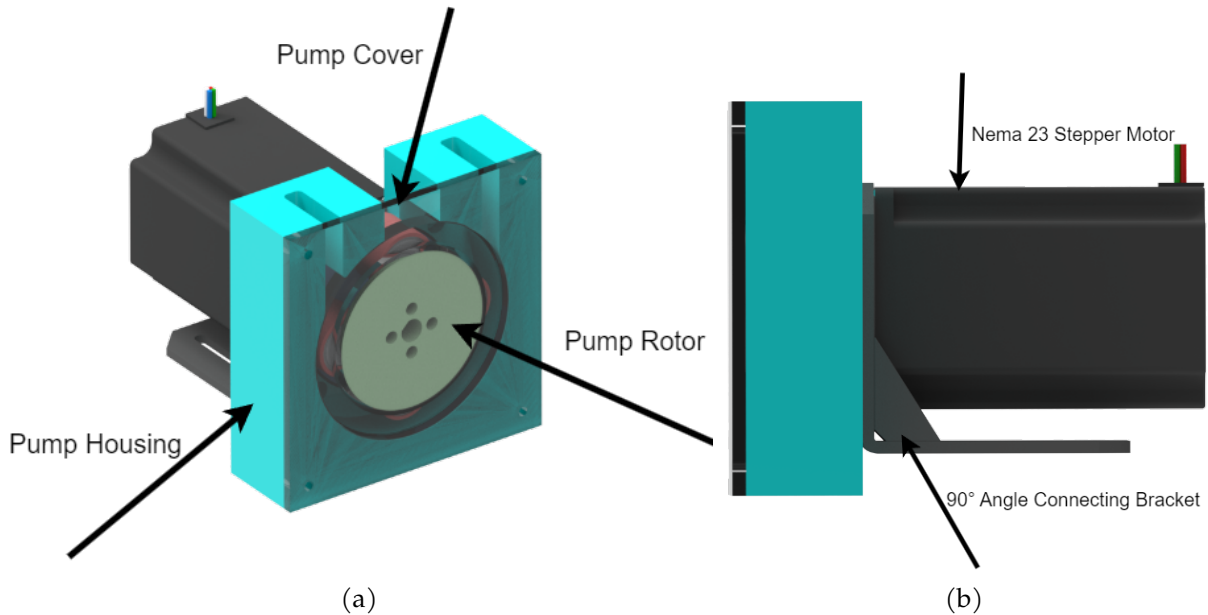


Figure 13: Isometric View of Peristaltic Pump, the Housing, Rotor and Cover are Indicated with Arrows (a), Side View of Peristaltic Pump, the Motor and 90° Angle Connecting Bracket are Indicated with Arrows (b)

The subsequent parts of the peristaltic pump are discussed further in the following sections

3.2.1 Pump Rotor

The rotor of the peristaltic pump is shown in Figure 14.

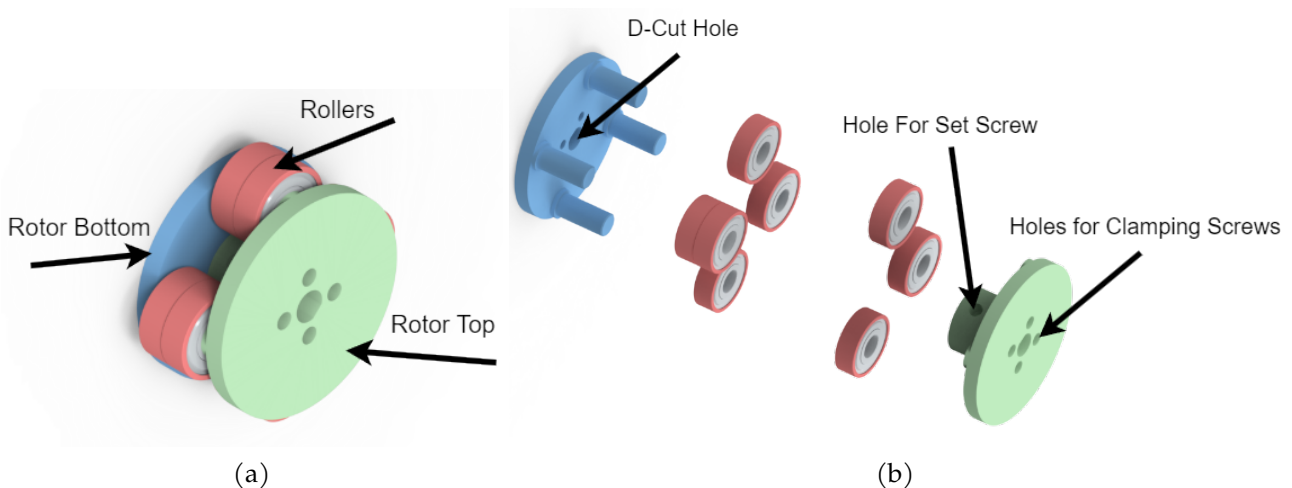


Figure 14: Isometric View of the Rotor with the Rollers (Red) Top (Green) and Bottom (Blue) Parts (a), Exploded View of the Rotor with Indications for the D-cut and Screw Holes (b)

The rotor is based on two pieces, top and bottom, that clamp together to hold a number of rollers in place. In Figure 14b it is shown how these parts interact: The bottom part (in blue) has protruding rods over which standard 608zz ball bearings [13] are installed. The top part (in green) slides over the rods and can be clamped with screws to the bottom part to hold the bearings in place. The bearings are clamped such that the inner race is fixed, while the outer race is free to move, due to a small bump

around the rods that only interact with inner race. To be able to screw into the different parts, some threaded inserts [14] were installed in the top part. In the middle of the two parts, a D-cut hole is made that can slide over the shaft of the stepper motor. The threaded inserts were also used here to be able to install a set screw that clamps the rotor to the shaft of the motor.

The rotor geometry was designed through a parametric framework, taking the number of rollers and rotor diameter as primary inputs and automatically adapting the design accordingly. These two parameters have the following influence on the behaviour of the pump:

- **Rotor Diameter:** A larger rotor diameter increases the moment arm and thus the torque demand on the motor. Additionally, a larger diameter means that for a single step, each roller moves a larger distance. This means that more fluid is displaced per step and thus reduces the accuracy.
- **Amount of Rollers:** A larger amount of rollers decreases the flow irregularity as it increases the pulsation frequency but reduces it's amplitude. This irregularity is inherent to the peristaltic principle: as a roller disengages, the tube expands again and the fluid stops begin pushed forward and the flow can even reverse [9].

To keep the achievable accuracy as high as possible, the rotor diameter was minimized and fixed as a parameter. The number of rollers was maximised to the amount that could be fitted on the rotor, in this case four, to have a more continuous flow.

3.2.2 Pump Housing

The housing of the Peristaltic pump is shown in Figure 15

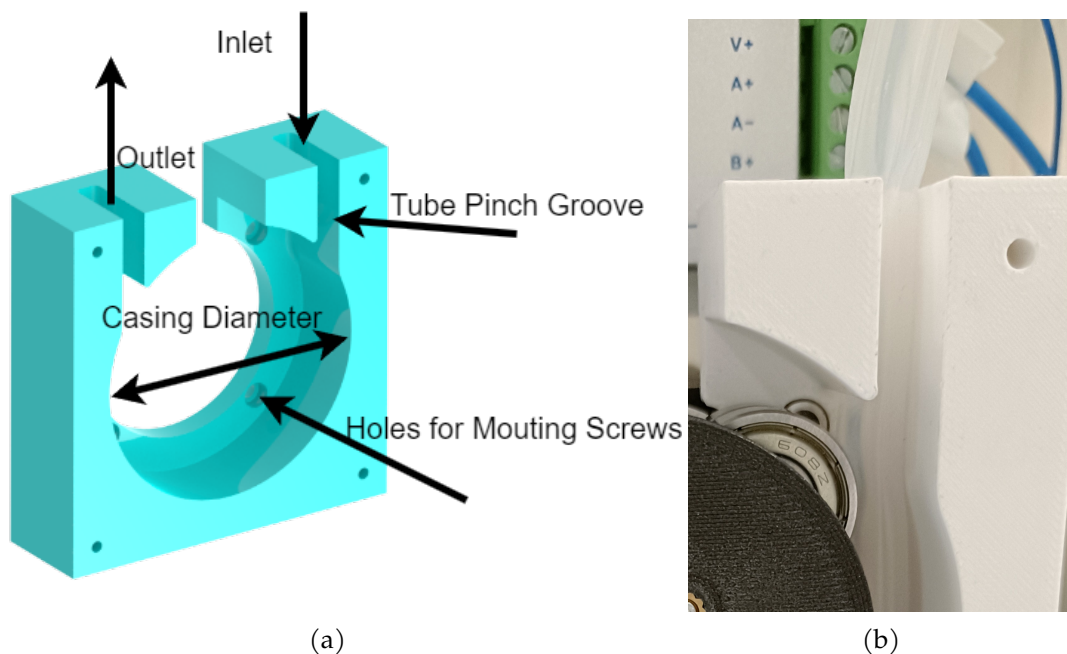


Figure 15: Isometric View of the Pump Housing. The Tube Pinch Groove, Diameter of the Inner Casing, Screw Holes and Flow Directions are Indicated with Arrows (a), Pinch Groove of the Housing with a Flexible Tube Installed (b)

The housing of the peristaltic pump is a single part with some distinct features. There are four screw holes to mount the housing to the stepper motor and the 90-degree bracket, see Figure 13. Furthermore, a pinch groove secures the tubing in place at both ends of the semi-circular cutout, preventing it from shifting during operation. This also enables the user to easily swap out the tubing if it needs to be replaced or maintenance needs to be performed, see Figure 15b.

Similarly to the rotor, the housing was designed through a parametric framework. In this case based on the dimensions of the full rotor as in Section 3.2.1 and some tube parameters: The inner and outer diameter of the tubing and the occlusion. The occlusion can be understood based on Figure 16.

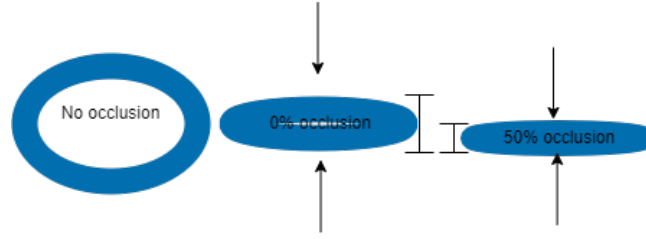


Figure 16: Illustration of the Occlusion, Representing the Amount of Compression of the Tube

The occlusion represents the level of compression the flexible tube is subjected to [10]. The occlusion percentage ϵ can be calculated as in Equation 1 using the inner and outer diameter of the tube d_{inner} and d_{outer} and the remaining thickness after compression in the pump R .

$$\epsilon = \frac{(d_{outer} - d_{inner}) - R}{(d_{outer} - d_{inner})} * 100\% \quad (1)$$

The occlusion mainly affects how well the tube seals against itself. This means that a higher occlusion allows for less leakage of the fluid between the rollers. A higher occlusion however is not always wanted since it creates more stress on the tube, leading to additional wear and thus a lower lifespan [10]. Since silicone is generally a very viscous fluid (for example, see [11]), backflow/leakage is not a real issue since a high viscosity means a greater resistance to flow. Therefore the occlusion is fixed at only 10% to maximize the lifespan and thus increase the ease of use.

The parametric framework for the housing is integrated in a Matlab script. The inner and outer diameter of the tube and the wanted occlusion factor can be entered and the output is the necessary diameter for the housing and the width of the pinch groove, see Appendix A. This means that different types of tube can easily be implemented, simply by printing a new housing

3.2.3 Pump Tubing

The flexible tubing is a critical component of the peristaltic pump, as several of its physical properties directly influence the pump's performance. The inner diameter, wall thickness, material stiffness, material composition... each affect different aspects of the pump behaviour [15, 9]. The most important properties are discussed below.

- **Tube Inner Diameter:** Increasing the inner diameter of the tube increases flow rate as more fluid is pushed through the tube per rotation. An increase in the diameter of the tube however leads to an increase in the irregularity of the flow [9]. This can be understood by looking at Figure 17.

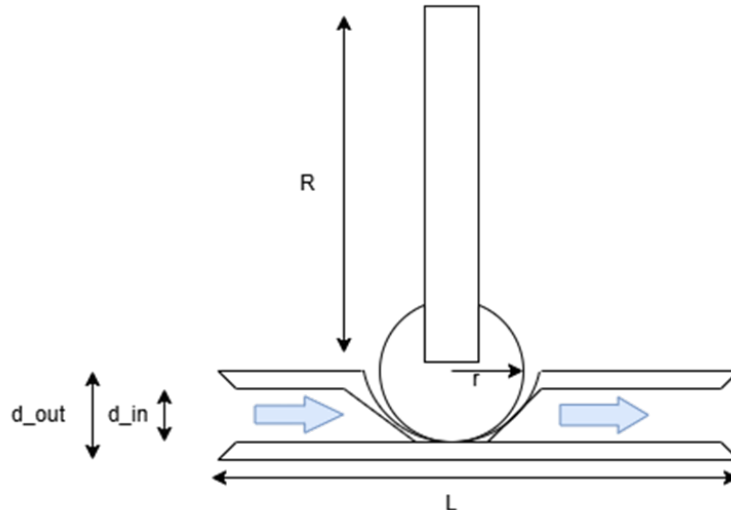


Figure 17: Illustration of the Effect of a Larger Tube Inner Diameter on the Flow Irregularity.

As the inner diameter of the tube increases, the volume each roller occupies increases. This results in a larger 'gap' when a roller disengages that needs to be filled by the fluid before it can move forward again and thus a larger pulsation or flow irregularity.

The most important influence of the inner diameter is the minimum displaced volume per rotation, or in this case per step of the Nema 23 stepper motor. A simplified model is considered to indicate if a certain inner diameter can comply with the required accuracy of $\pm 0.1g$. This model assumes that the tube is filled perfectly with fluid, that there is no backflow between the rollers and a density of silicone $\rho_{silicone} \approx 1.03 \frac{g}{cm^3}$ [11]. The inputs are the tube diameter d_{in} , the step size θ and the radius R of the rotor.

$$\begin{aligned} InnerSurfaceTube &= S_{tube} = \frac{\pi \times d_{in}^2}{4} \\ Volume/step &= R \times \frac{2\pi \times \theta}{360^\circ} \times S_{tube} \\ Mass/step &= Volume/step \times \rho \end{aligned} \quad (2)$$

This model indicates a quadratic relationship between the minimum dispensed volume and the inner diameter of the tube. Since very precise fluid control is required, this inner diameter should be minimized, however this is limited by other factors such as the viscosity of the silicone and the dispensing time, see later.

- **Tube Thickness:** An increased tube thickness means that the torque required to rotate the rotor is increased as more energy is needed to compress the tube. This also means that the tube has a higher 'spring back' or faster recovery time after a roller has passed and compressed the tube. This means the inner opening of the tube is quickly restored and the flow is less constricted to fill the tube again, resulting in a better filling of the tube and higher average flow rate.
- **Tube Stiffness:** The impact of the tube stiffness is approximately the same as the thickness. Some preliminary testing indicated that the stiffness and thickness of the tube have a large influence on the performance on the tube, see Figure 18

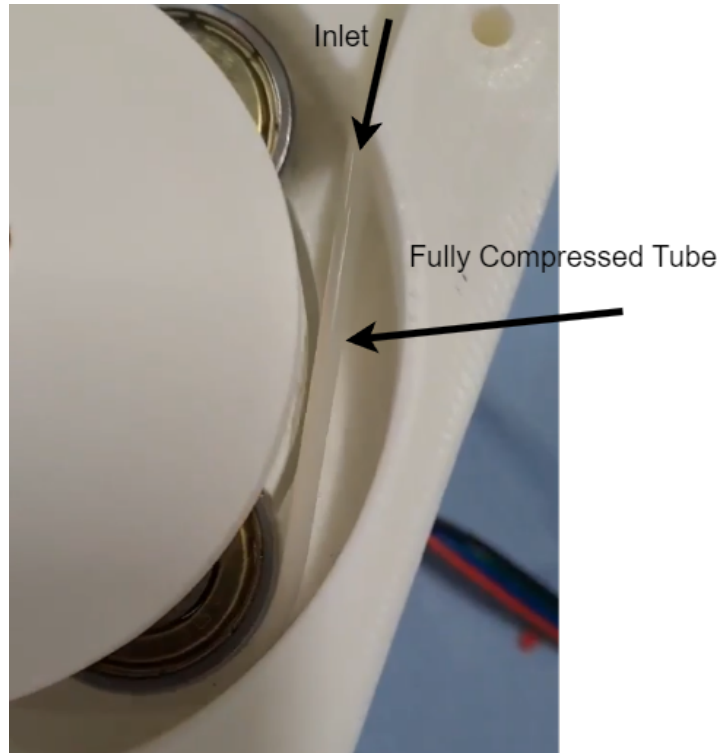


Figure 18: Preliminary Test with Silicone Tube with an Internal Diameter of 3mm, Wall Thickness of 1mm and Hardness 60 ShA: The Tube Stays Fully Collapsed After Rollers Move over it due to Large Viscous Adhesive Forces of the Silicone.

The test in Figure 18 was performed using a flexible silicone tube with an internal diameter of 3mm, wall thickness of 1mm and a Shore A Hardness of 60 [16]. It can be observed that this combination results in almost zero flow: The tube is too weak to expand again after the roller moves over it due to the viscous adhesive forces of the Silicone. Additionally the inner diameter of the tube is too small to allow easy flow of the silicone into the tube to expand it.

- **Tube Material:** It is important for the tube to be chemically compatible with the fluid. In this case this means that the material of the tube should not cause any (partial) hardening of the silicone components, especially in the tube.

During testing, it became apparent that the silicone tubing used up to this point [16], due to its ready availability, caused a chemical reaction with the silicone components, leading to premature curing of the components inside the tube. This can be seen in Figure 19

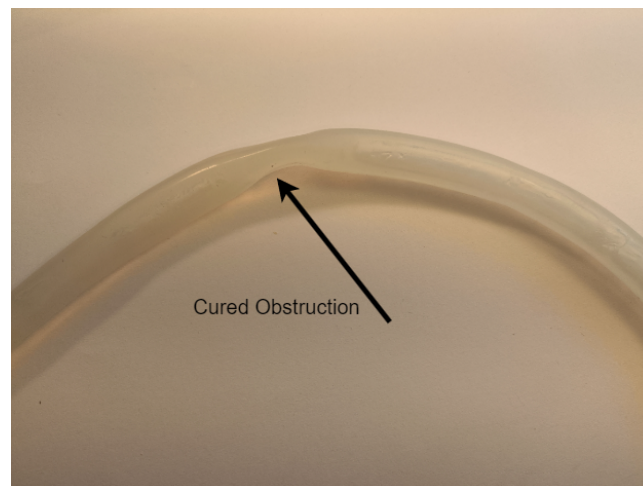


Figure 19: Silicone Tube with an Obstruction due to a Chemical Reaction Causing the Pumped Silicone Component to Cure Inside and Fuse the Tube Together.

Given the influence of these parameters, the following balances were made. The inner diameter of the tube was minimised as much as possible to increase the achievable accuracy, but due to the high viscosity and thus high resistance to flow, this had to be traded off against the maximum allowable dispensing time. A larger inner diameter was therefore required to dispense the necessary volume within the time constraint, while also ensuring the tube does not stay collapsed due to too slow filling of the tube at the inlet side such as in Figure 18. The hardness and thickness of the tube were then maximised (given the availability of flexible tubes) to increase the ability to expand again and ensure good filling of the tube. A balance was found with a 60 ShA silicone tube with an inner diameter of 5mm and an outer diameter of 10mm. For reference, the datasheet of the silicone tubing is included in Appendix B

This type of tubing, being made from silicone itself, causes the silicone components to cure prematurely inside the tube, as illustrated in Figure 19. An alternative was therefore sought and identified in consultation with Tubes International [17], however, due to delivery lead times, this alternative could not be tested within the scope of this project. Although the material composition differs entirely (polyolefin), the shore hardness and dimensions are comparable, suggesting that the alternative tubing should function correctly. Implementation would require only a new housing to be printed, adapted to the inner and outer diameter of the new tubing, as described in Section 3.2.2. The datasheet of the alternative tubing is included in Appendix C.

3.2.4 Connections and Restrictions

To interface with the container for the silicone and the valve, some connectors were used. They also play a role in the overall performance of the pumping system. Minimising flow restrictions throughout the system reduces the pressure drop across the tubing, which improves tube filling on the inlet side and facilitates fluid expulsion on the outlet side. This allows a smaller inner tube diameter to be used, which simultaneously increases dispensing accuracy as discussed above and reduces the minimum dispensable volume. To achieve this, several optimisations were made on both the inlet and outlet side of the system, which are discussed below.

- **Inlet at the Silicone Container:** The silicone is stored in a container above the pumps, see Figure 20 and Figure 11 for an overview.

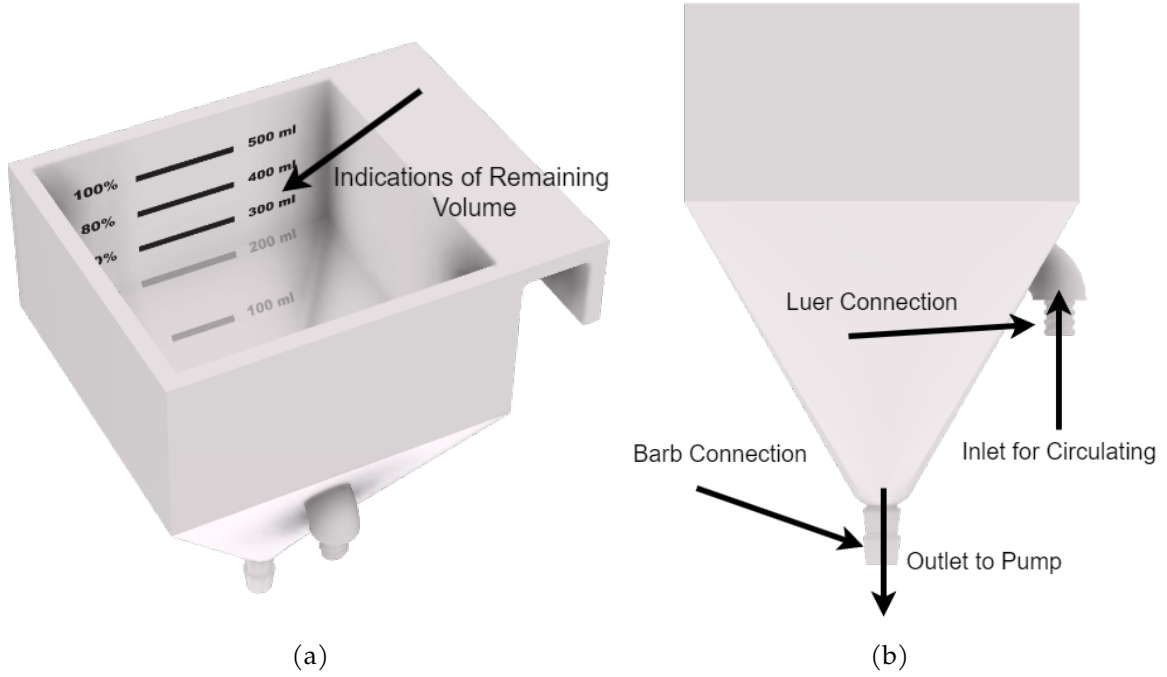


Figure 20: Isometric View of Silicone Container with Indicated Remaining Volume (a), Front View of Silicone Container with Indicated Connections and Fluid Flow Direction (b)

To improve ease of use, the container volume slightly exceeds the standard volume of a commercial silicone cartridge [11], allowing an entire cartridge to be dispensed at once. Volume markings are incorporated on the inner surface to indicate the remaining content. The container is designed to be printed as a single part without support material, enabling straightforward replacement in case of damage or when switching between different silicone types.

To connect to the tube of the pump, a barb fitting is made on the underside of the container. To understand the influence of the connections on the pressure drop and flow in the tube, the energy equation for internal pipe flow is considered in Equation 3 [18, §8.6]

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V}_1^2}{2} + gz_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V}_2^2}{2} + gz_2\right) = h_{lt} \quad (3)$$

In this equation, p_1 & p_2 represent the pressure at the start and end of the considered flow, $\bar{V}_1$ & $\bar{V}_2$ represent the average velocity across the pipe area and z_1 & z_2 the elevation. Furthermore, ρ represents the density, g the gravitational constant and α_1 & α_2 kinetic energy correction factors, which account for the non-uniform velocity profile across the cross-section; in practice α is often taken equal to unity. Finally, h_{lt} represents the total losses in the system, namely the major losses due to friction and the minor losses due to changes in direction, area or fittings.

The diameter of the hose remains constant from the container to the pump and thus $\bar{V}_1 = \bar{V}_2$ and since the container is higher than the pump, see Figure 11: $z_1 - z_2 > 0$ and remains constant. In this way Equation 3 can be rewritten as:

$$\frac{p_1 - p_2}{\rho} = h_{lt} - g(z_1 - z_2) \quad (4)$$

Equation 4 shows that an increase in friction or minor losses requires a greater pressure difference to maintain the same flow rate. Part of this pressure difference is naturally provided by the elevation head $g(z_1 - z_2)$, which acts as a gravity assist. It follows that minimising the losses h_{lt} and maximising the height difference reduces the required pressure drop and consequently promotes higher flow rates. This is the rationale behind two design choices: the barb fitting is made equal in diameter to the tube, avoiding any additional flow restriction (minor losses), and the container is positioned above the pump to exploit the hydrostatic pressure contribution.

- **Restrictor at the Outlet of the Pump:** At the final output, a small tube diameter is desired to allow precise dispensing of silicone in small droplets. This requires a reduction in internal diameter somewhere along the flow path. Since a valve is already present between the pump and the output (see Section 3.3, with an internal diameter of 3mm, this reduced diameter is maintained from the valve onwards to the output. Additionally, Luer connectors use an internal diameter of $\approx 3mm$ and are a quick way to connect tubing together. This approach avoids introducing additional flow restrictions and minimises further minor losses, while still achieving the required dispensing precision.

Considering Equation 3 once more, but now with $\bar{V}_1 \neq \bar{V}_2$ due to the change in cross-sectional area, the same principle applies: for a constant flow rate, both $\bar{V}_1$ and $\bar{V}_2$ are fixed, and any reduction in the total head loss h_{lt} directly reduces the pressure difference required to sustain that flow. Since the tube is flexible, it will expand slightly when the pressure increases. This negatively impacts the flow irregularity: when a roller disengages from the tube, not only will the silicone try to rush back to fill the void, it is pushed back by the residual pressure in the tube, increasing the irregularity. To minimize this effect as much as possible, a gradual restrictor was made to go from the 5mm to 3mm internal diameter to reduce the minor losses and thus reduce the required pressure, see Figure 21.

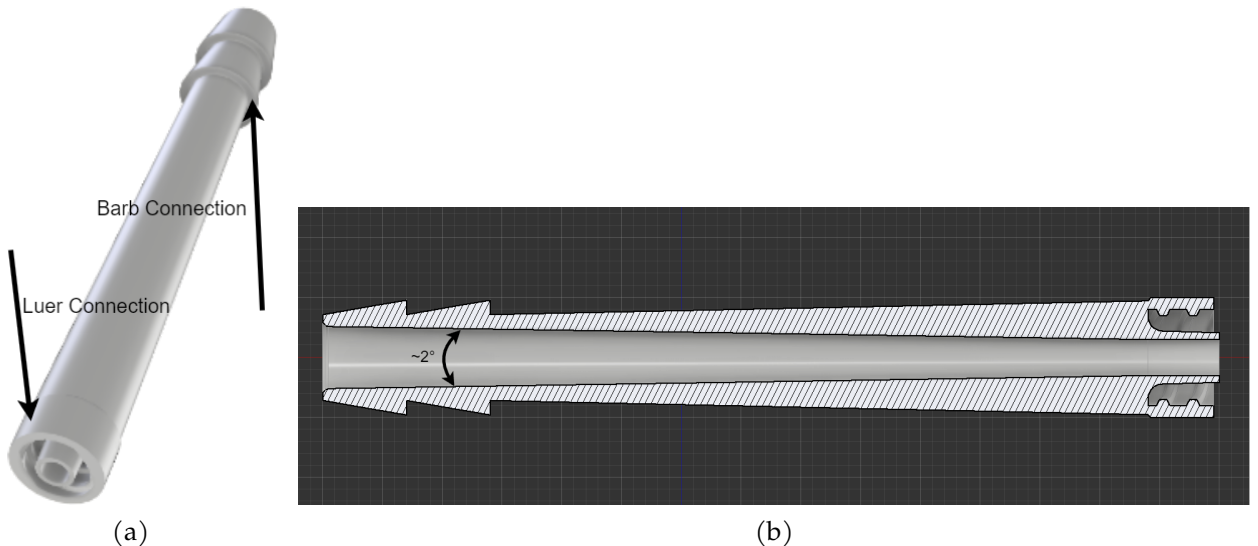


Figure 21: Isometric View of Gradual Restrictor to go from 5mm to 3mm Internal Diameter with Barb Connection for Pump Tubing and Luer Connection for Output Tubing (a), Internal Section of the Gradual Restrictor with Indicated Angle of 2° (b)

The minor losses h_{lm} due to restrictions can be calculated as in Equation 5 [18, §8.7]:

$$h_{lm} = K \frac{\bar{V}^2}{2} \quad (5)$$

In this equation $\bar{V}$ represents the mean velocity in the pipe in the smallest section (for a restriction after the contraction) and K represents the loss coefficient. As can be seen from Table 8.3 in [18, §8.7], for a certain area change a decrease in the included angle of a fitting results in a decreased loss coefficient. This means that there will be less pressure required to push the fluid through the valves and tubes, reducing the flow irregularity.

In summary, the design of the pumping system is the result of a series of interconnected trade-offs. The rotor diameter and number of rollers were selected to balance dispensing accuracy and flow continuity. The tubing parameters, namely inner diameter, wall thickness and hardness, were chosen to ensure adequate flow rates within the time constraints while maintaining sufficient tube recovery after compression. The connections and restrictions were optimised to minimise pressure losses throughout the system, reducing flow irregularity and maximising dispensing accuracy. The container was designed for ease of use, allowing a full silicone cartridge to be dispensed at once while providing a gravity-assisted inlet. Finally, the chemical compatibility issue identified with silicone tubing has been addressed by identifying a possible alternative.

3.3 Design Valve Unit

The main objective of the valve unit is to add the ability to switch from dispensing mode to circulation mode. The valve unit has three ports. One input port, and two output ports. One output port feeds the silicone back to the container. The other output port feeds the silicone to the dispensing unit. A general overview of the unit is shown Figure 22

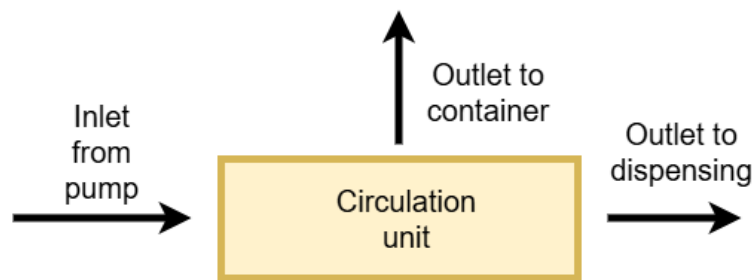


Figure 22: Illustration of the Valve Unit: 1 Input Port, Two Output Ports

There is a wide range of different silicone that is used in the lab, with different properties. A fundamental advantage of the design is the possibility to quickly swap between components by changing the tubing and container. To avoid contamination of one kind of silicone on another would mean that the entire circulation unit should be replaced. For this reason, the development of a customized design is more beneficial. The custom valve unit is designed so that only one cheap component has to be replaced (the stop-cock valve), instead of the whole part. The custom designed parts are 3D printed out of PLA. This material is chosen because it is easy available, cheap and has a high enough strength for this application. Although they are not standard components, they are still small and simple components that don't take long to print and do not need to be reprinted when a valve is changed.

The most important requirements and boundary conditions for the design of the valve unit are repeated here:

- **Circulating:** The valve unit adds the ability for the machine to circulate the silicone back to the container.
- **Repairing:** The valve unit is easy repairable, with standard components.
- **Interchangeability :** The valve system can be adjusted to a different kind of silicone.

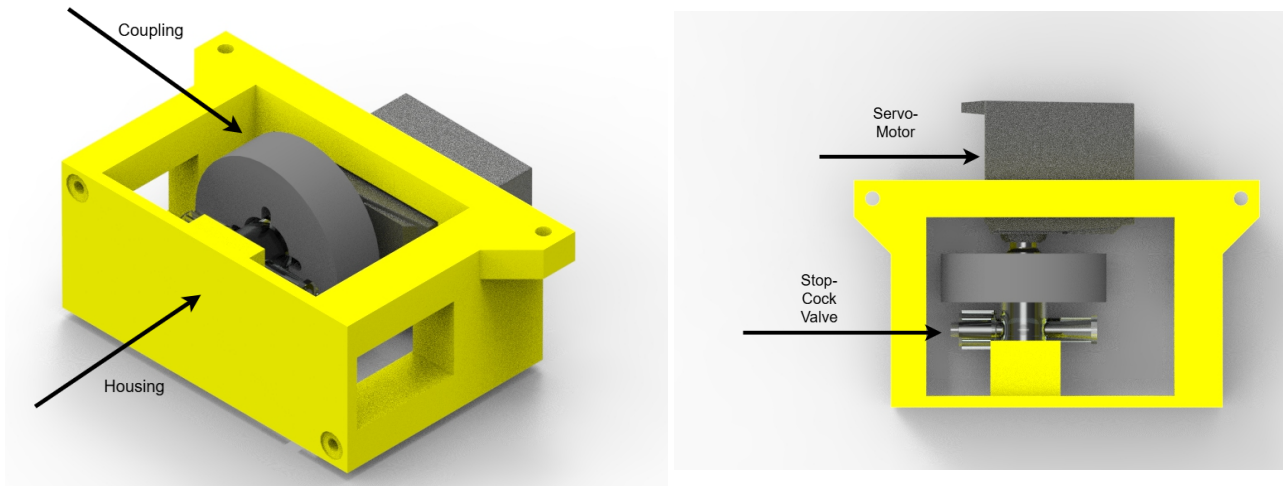


Figure 23: Isometric View of Circulation Unit, the Coupling and Housing are indicated with Arrows (a), Top View of Circulation Unit, the Servomotor and Stopcock Valve are Indicated with Arrows (b)

The unit is constructed out of 4 parts: the valve, the coupling, the actuator and the housing. The whole part is displayed in Figure 23. The unit works by activating the servomotor. The handle of the stopcock valve is turned with the help of the coupling. The motion of the body of the stopcock valve is restrained because it is geometrically locked by the top. The new position of the valve blocks one of the outlets so the silicone is redirected to the other one.

The main parameter influencing the design of the is the static friction of the valve. The static friction is influenced by the size of the valve, the silicone type and the time the silicone has been inside the valve (and the valve has not moved). The valve unit can be improved by quantitatively determining the influence of the silicone on the static friction of the valve. In this way, a more accurate trade-off in the size of the valve can be made. In this way the required torque can be calculated more accurately and an appropriate motor can be chosen.

The working principle and design choices of each component are explained in more detail in the following subsections.

3.3.1 Valve

A stopcock valve is chosen to switch between circulation and dispensing line. This kind is chosen because it is cheap and commercially available [19]. It is easy controllable by the handle. It is also convenient to connect luer-locks and other attachments. The used valve can be seen in 24.

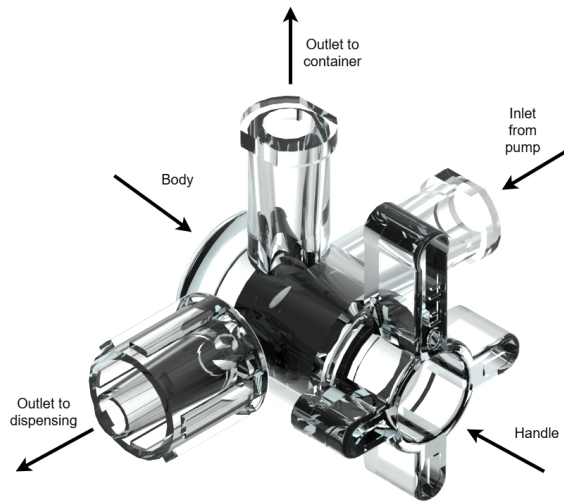


Figure 24: Isometric View of Stopcock, Inlet from Pump, Outlet to Container, Outlet to Dispensing, Handle and Body are Indicated with Arrows

3.3.2 Actuator

A servomotor is chosen to actuate the unit. This is because they are cheap and easy controllable. They are small and can very accurately move to a certain position. They are also widely commercially available [20]. The used motor can be seen in Figure 25.

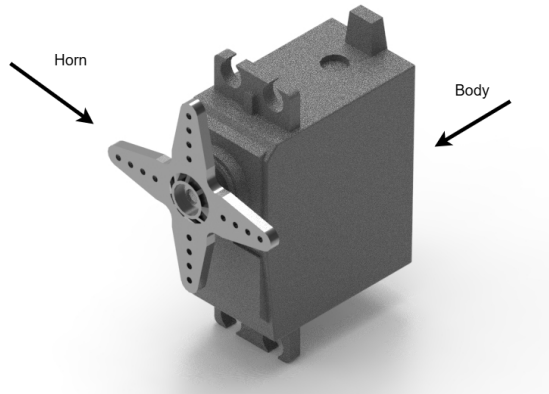


Figure 25: Isometric View of Servomotor, Horn and Body are Indicated with Arrows

The choice of the servomotor is a trade-off between size and power consumption/price. The power and torque needed to overcome the static friction is directly related to the size of the valve. A large/powerful servomotor is beneficial but gets fined with a higher cost and larger total size and weight of the unit. An increase in current is another way to increase the torque. The datasheet for the servomotor is added in Appendix D

3.3.3 Coupling

To transfer the motion of the servomotor to the valve, a coupling is needed. The coupling is a custom design, to fit the shape of the servo horns and the handle of the valve. The Coupling can be seen in Figure 26.

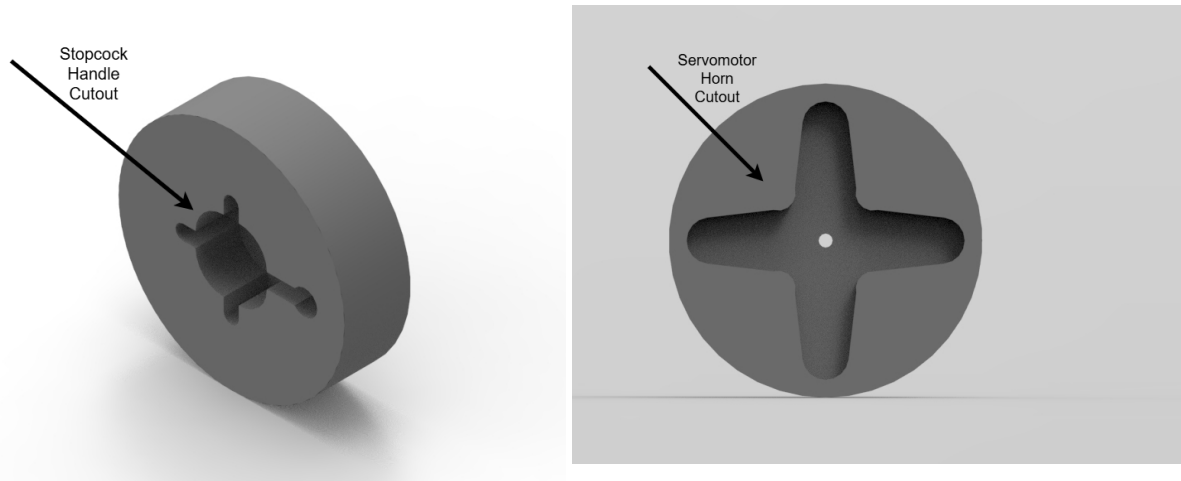


Figure 26: Isometric View of the Coupling, the Cutout of the Handle of the Stopcock Valve is Indicated with an Arrow (a), Back View of the Coupling, the Cutout of the Servomotor Horn is Indicated with an Arrow(b)

3.3.4 Housing

The housing is the part that holds the whole unit together. It is also used to mount the valve unit to the frame. It is a custom design and the same material as the coupling for the same reasons as above. The two parts of the housing can be seen in Figure 27.

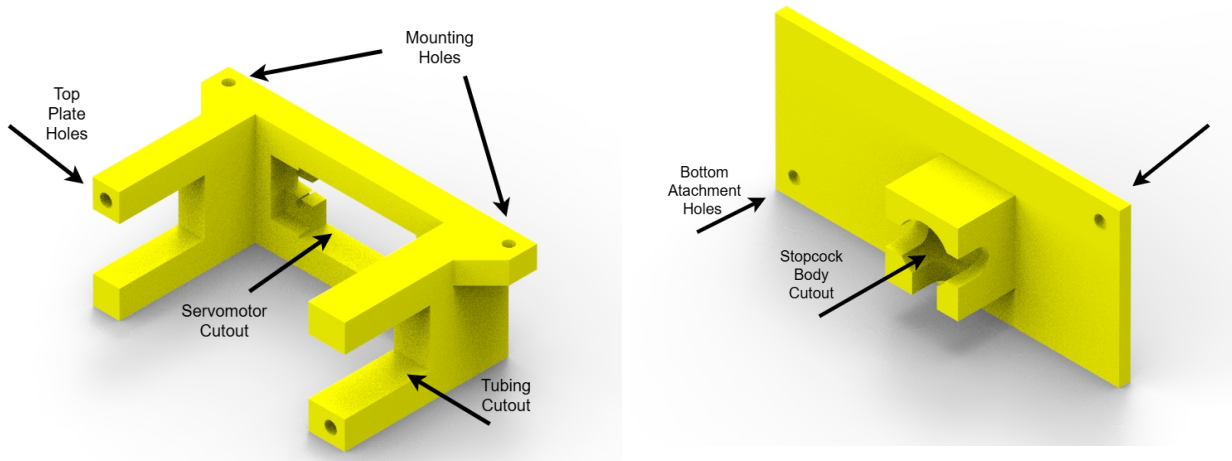


Figure 27: Isometric View of the Bottom of the Housing , the Holes for the Bolts, the Cutout for the Servomotor, The Cutouts for the Tubes are Indicated with an Arrow (a), Isometric View of the Top of the Housing, the Holes for the Bolts, the Cutout for the Body of the Stopcock are Indicated with Arrows(b)

3.4 Design Frame

The frame is built up of aluminium profiles [21], because they provide sufficient strength, are commercially available and relatively cheap. The profiles used are easy connectable via connection angles. The designed units discussed before are mounted on the profiles with nuts and bolts, see Figure 1.

3.5 Design Electronics

The design of the electronics was a direct result of the chosen electrical components, and these were each chosen based on the requirements and the design choices made in the value analysis.

3.5.1 Stepper motors

The stepper motors are controlled through stepper drivers, as shown in Figure 28. The motors are powered directly from the 24 V power supply through the driver, while the Raspberry Pi 4 (RPI4) [22] only provides the control signals for pulse, direction, and motor enable/micro step selection.

The driver inputs are connected using a common-positive configuration. This means that the positive input terminals, PU+, DR+, and MF+, are connected to 5 V. The RPI4 [22] GPIO pins are connected to the negative input terminals, PU-, DR-, and MF-, as listed in Table 2. This is used because the RPI4[22] GPIO output voltage is 3.3 V, which is lower than the 5 V input side of the driver.

Table 2: Stepper Driver Wiring to the RPI4 [22] Using Common-Positive Driver Inputs and BOARD Pin Numbering.

Driver terminal	Raspberry Pi 4 [22] connection	Motor
PU-	Physical pin 7, GPIO4	1
PU-	Physical pin 11, GPIO17	2
PU-	Physical pin 13, GPIO27	3
PU-	Physical pin 15, GPIO22	4
DR-	Physical pin 18, GPIO24	all
MF-	Physical pin 16, GPIO23	all
PU+	5 V	all
DR+	5 V	all
MF+	5 V	all

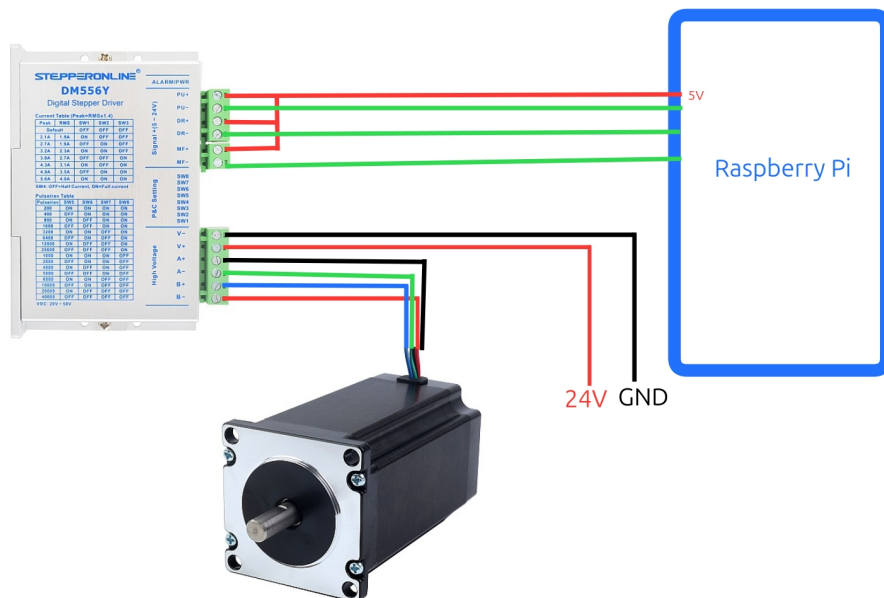


Figure 28: Wiring of Stepper Motor and Driver

3.5.2 Servo motors

The servo motors are powered with 5 V from an TRACOPOWER TSR 2N buck converter, using the main power supply as input. This is necessary because the RPI4 [22] cannot provide enough current to power the servos on its own. The servo and wiring are shown in Figure 29, and the RPI4 [22] connections are listed in Table 3.

The selected servo motors have enough torque to turn the valves during normal operation. However, when the valves have been idle for more than a day, they can lock up and require a larger torque to start moving again. For this reason, a back-and-forth valve movement is included in the automatic circulation routine to keep the valves from sticking.

Table 3: Servo Wiring to the RPI4 [22] Using BOARD Pin Numbering.

Servo	Servo wire	Raspberry Pi 4 connection
Servo 1	Signal	Physical pin 12, GPIO18
Servo 2	Signal	Physical pin 32, GPIO12
Servo 3	Signal	Physical pin 35, GPIO19
Servo 4	Signal	Physical pin 33, GPIO13
All servos	VCC	External 5–6 V servo supply
All servos	GND	Raspberry Pi 4 [22] GND and servo supply GND

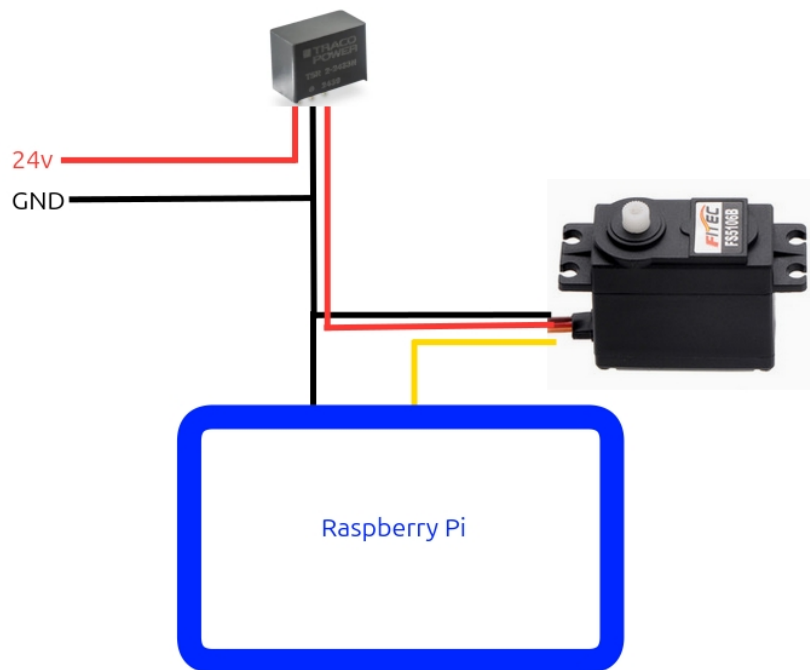


Figure 29: Wiring of the Servo Motor

3.5.3 Load cell

The machine has to measure the dispensed mass accurately, because this value is used to decide how much silicone still has to be dispensed. The final dispensed mass has to be within 0.1 g of the target value. For this reason, a 200 g single point load cell is placed under the container [23].

The selected load cell has a combined error of $\pm 0.02\%$ of its nominal capacity, which corresponds to approximately ± 0.04 g for the 200 g version. This is accurate enough for the required dispensing precision.

Because the signal from a load cell is very small, it cannot be read directly by the RPI4 [22]. Therefore, an HX711 [24] load cell amplifier is used. The HX711 [24] is a 24-bit analog-to-digital converter designed for weighing applications.

The combination of the load cell and HX711 [24] makes it possible to measure small changes in dispensed mass with a simple digital interface. The load cell and HX711 [24] are powered by an LDO, which converts 24 V into 5 V. This keeps the supply voltage stable and reduces noise from stepper or servo current peaks. The load cell and HX711 wiring is shown in Figure 30, and the RPI4 [22] connections are listed in Table 4.

Table 4: HX711 wiring to the RPI4 [22] using BOARD pin numbering.

HX711 pin	Raspberry Pi 4 [22] connection	Signal
DOUT / DT	Physical pin 29, GPIO5	Data output
PD_SCK / SCK	Physical pin 31, GPIO6	Serial clock
VCC	5 V from LDO	Supply voltage
GND	Raspberry Pi 4 [22] GND and LDO GND	Common ground

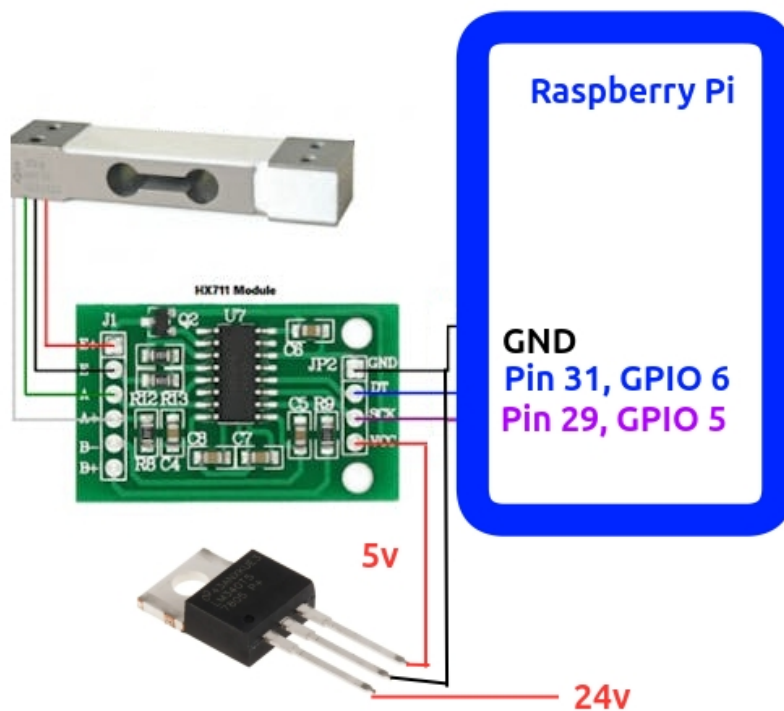


Figure 30: Wiring of the Load Cell and HX711 Amplifier

3.5.4 Raspberry pi

A Raspberry Pi 4 (RPI4) [22] is used as the central controller of the machine, as shown in Figure 31. It was chosen because it can run the graphical user interface while still providing GPIO pins to communicate with the different electronic components. This makes it possible to combine the user input, control algorithm, motor control and load cell readout on one platform.

The main program runs on the RPI4 [22]. Based on the selected mode, the RPI4 [22] sends control signals to the pump motors and the valve servomotors. It also reads the digital signal from the HX711 [24] load cell amplifier to determine the dispensed mass.

The motors are not powered directly from the RPI4 [22], because the GPIO pins can only provide low-current 3.3V control signals. Therefore, the stepper motors are controlled through external stepper drivers and the servomotors are powered separately. The RPI4 [22] only provides the control signals, while the motor drivers and power supply provide the required current.

Using a RPI4 [22] makes the system flexible, because changes to the interface, dispensing algorithm or calibration procedure can be implemented in software using git pull, without changing the hardware.

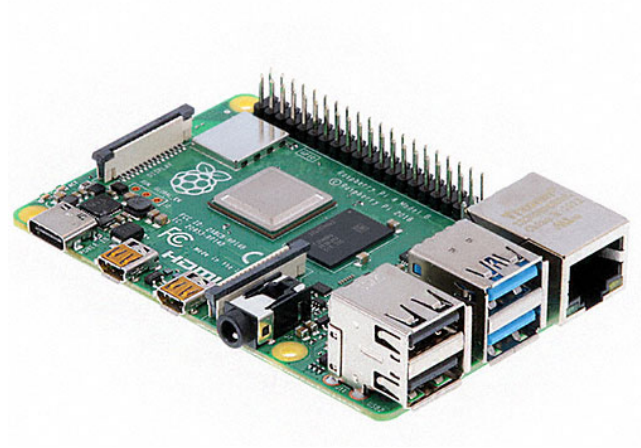


Figure 31: Raspberry Pi 4 Used as the Central Controller [22]

3.6 Design Interface

This section will discuss, how the user can use the interface to obtain results. Section 3.6.1 shows how the interface is made and section 3.6.2 explains how the user should use the interface and how to start it.

3.6.1 Interface structure

The interface was made in pygame and has been made as intuitive as possible. Figure 32 shows a flowchart between the different screens of the program, the purple lines are lines where a return button is available to go to the previous screen and the black lines only go forward.

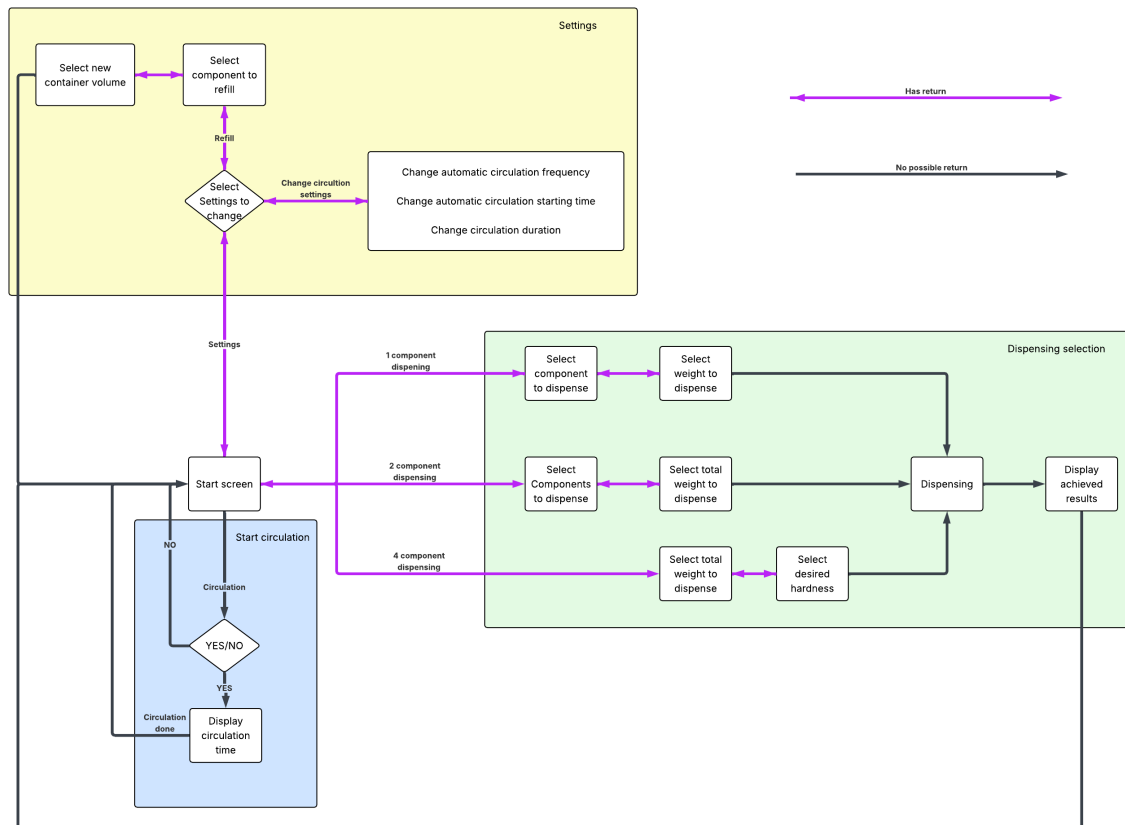


Figure 32: Flowchart Screen Navigation

3.6.2 Interface guide

When the machine boots up, a Linux-style desktop environment is shown. The user can navigate this environment using the touchscreen and the on-screen keyboard.

To start the machine UI, open a terminal and go to the Interface folder. Then pull the latest software from Git if needed, and run the `Interface_rpi.py` file.

Listing 1: Starting the Raspberry Pi interface using CLI

```
1 cd Silicone_dispenser
2 git pull      # Only necessary if software changes were made
3 cd Interface
4 python Interface_rpi.py
```

After running these commands, the user interface starts.

Interface homepage:

On most pages in the interface, the top bar shows how much silicone is left in each bucket. This allows the machine to reject a request if there is not enough silicone available. Each time the silicone is refilled, the user should update this in the settings, see section 3.6.2.

On the homepage of the interface, seen in Figure 33, the user can choose to dispense just one component, 2 components with the same mass each, or 4 components with a desired hardness they choose. The user can also circulate all the silicone in the system, or go to settings where they can schedule automatic circulation, and log refilling of the buckets.

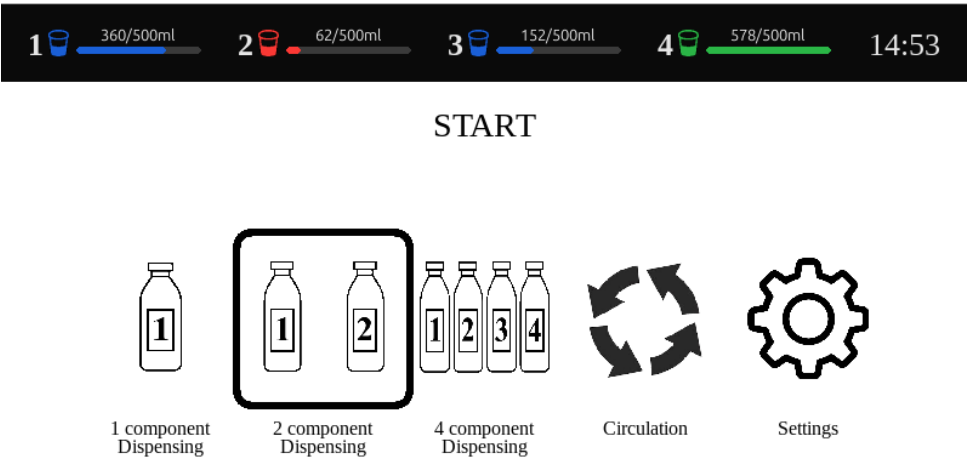


Figure 33: Interface on the Home Page

1 component dispensing When the user chooses to dispense one component see Figure 34, they then choose from which bucket they want to dispense, and then select the desired weight. After this, the dispensing will start, the progress will be displayed on the screen.

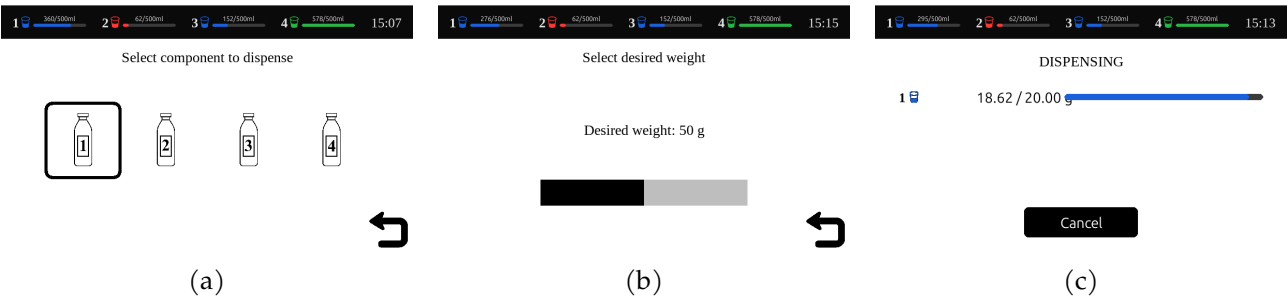


Figure 34: One Component Dispensing Screen: Bucket Selection (a), Mass Selection (b), Dispense Progress (c)

2 component dispensing:

Two component dispensing works similarly to single component dispensing, but the user first has to choose which two of the four buckets they would like to select from. Then user selects how much they want to dispense in total, half of that mount will be dispensed from both part A and part B of the chosen silicone to have the right ratio to cure.

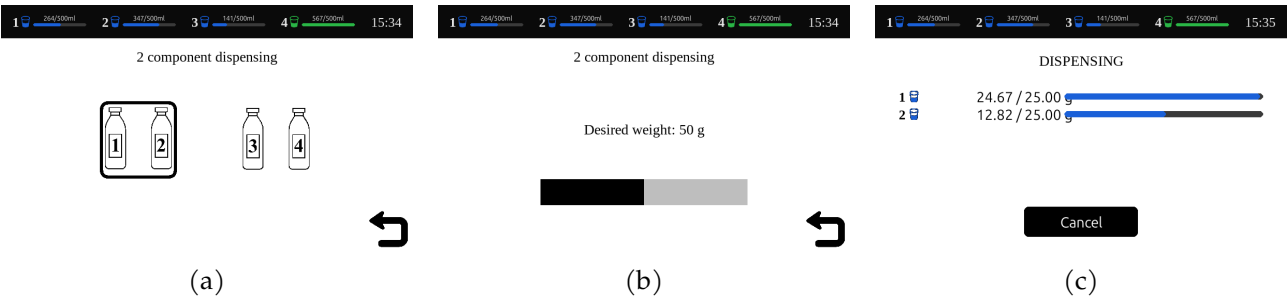


Figure 35: Two Component Dispensing Screen: Bucket Selection (a), Mass Selection (b), Dispense Progress (c)

4 component dispensing:

Four component dispensing asks the user for the desired total mass as in Figure 37a, and the desired harness as in Figure 37b. The desired ratio is calculated using an interpolation from the data of Figure 36.

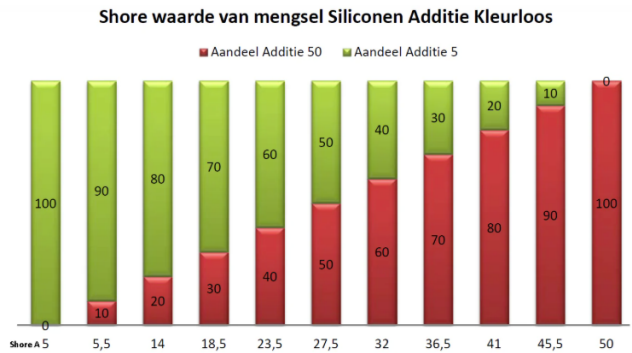


Figure 36: Required Ratio of Silicone Types for Multiple Hardnesses [25]

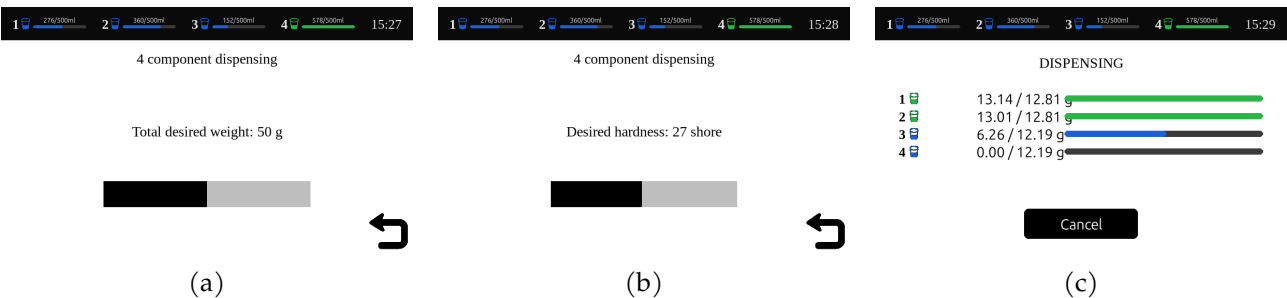


Figure 37: Four Component Dispensing Screen: Weight Selection (a), Hardness Selection (b), Dispense Progress (c)

Circulation settings:

When the user chooses Circulation settings (Figure 38a), a form (Figure 38b) is presented where the user can select the circulation frequency (daily or weekly), circulation time (day of the week and time of circulation), and circulation duration (amount of minutes it circulates per bucket).

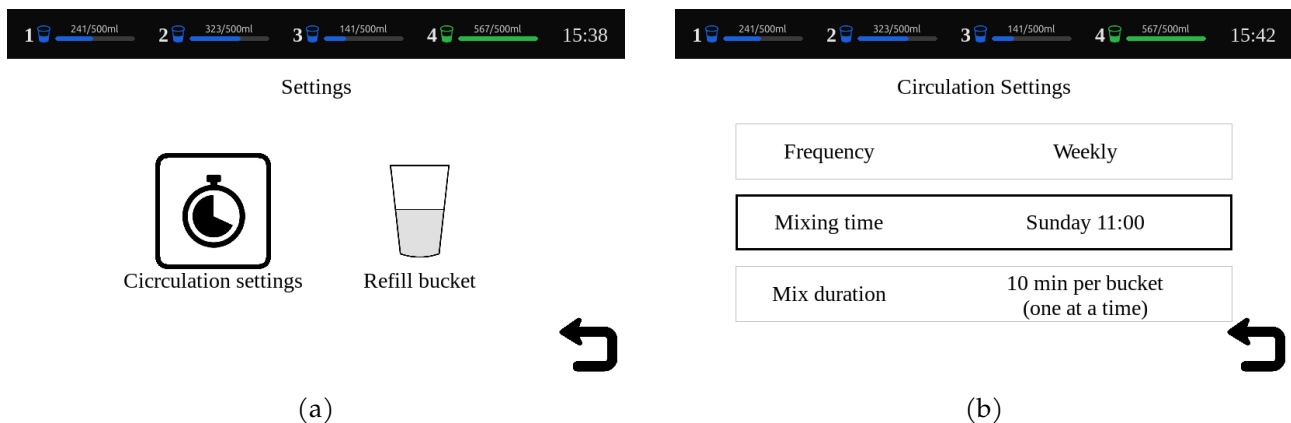


Figure 38: Circulation Settings: Circulation Settings Option (a), Circulation Settings Form (b)

In the settings the user can change the automatic circulation settings:

- Automatic circulation frequency (daily, weekly)
- Automatic circulation starting time (day of the week, time)
- circulation duration

Refill bucket:

When the user refills a bucket with silicone, this should be logged in the machine. This option is located in the settings, see Figure 39a. The user selects the bucket they want to refill, see Figure 39b, and select the new total volume of silicone that is in the bucket after refilling, see Figure 39c.

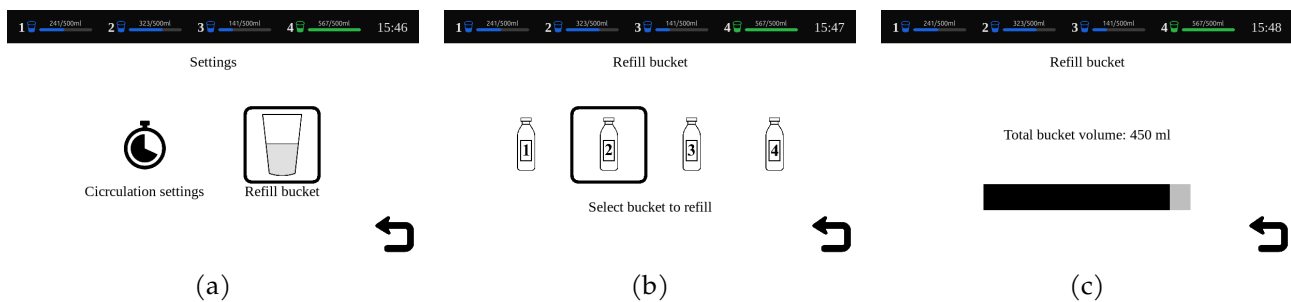


Figure 39: Refill settings: Refill Option (a), Bucket selection (b), New volume selection (c)

The refill section contains:

- Select container to refill
- Select new total volume in container

All these settings (circulation and refill) are stored in a file (saved_settings.json) so they are not lost when the Raspberry pi is restarted.

Circulation:

If the user selects to do Circulation now, the program will take the circulation duration from saved_settings.json and circulate for this time. The interface for this can be seen in Figure 40

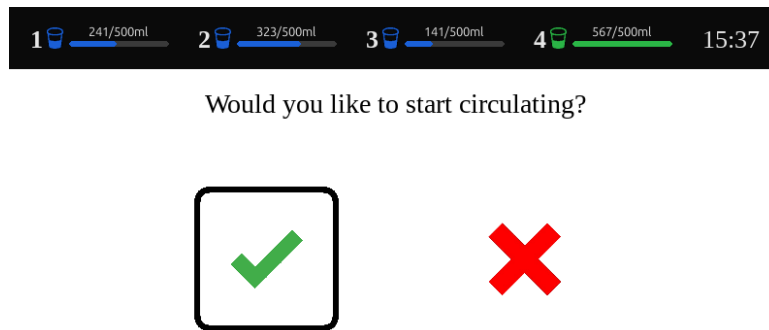


Figure 40: Circulate

3.7 Design of Control

The control is made using three discrete parts and their constraints. Section 3.7.1 explains how the pump was modelled, Section 3.7.2 shows the actual control algorithm and Section 3.7.3 explains how load cell was used.

3.7.1 Pump modelling

The steps/g ratio is calculated in multiple steps:

The weight is converted to a volume ($\rho = 1.06 \frac{g}{cm^3}$).

The peristaltic pump is modelled as a stiff tube where an infinitesimally small plane pushes the liquid forward. This results in following equation for Volume displacement:

$$V(x) = \pi * r^2 * x \quad (6)$$

The pump has a circular arm that pushes the liquid forward, such as in Figure 17:

$$x(step) = l_{arm} * 2 * \pi / (steps/revolution) * steps \quad (7)$$

$$V(step) = \pi * r^2 * l_{arm} * 2 * \pi / (steps/revolution) * steps \quad (8)$$

$$m(step) = V(step) * \rho \quad (9)$$

This model is quite inaccurate because:

- The tube is not infinitely stiff and will expand
- There will be backflow => not all the volume in a part is pushed forward
- The fill rate is not 100%
- The rollers take space away where the liquid would be

There were two options to improve this design:

- A more complex model that takes into account:
 - Tube stiffness
 - Pressure gradient in tube
- Adding an empirically found error factor

For now the only the second option was used. But as some variables change with time and others are pump specific, one test doesn't suffice for this error coefficient. Instead every time the system dispenses, after the first (and largest) dispense, the ratio between the expected and the dispensed weight and is taken as the error coefficient.

3.7.2 Control procedure

In this control strategy, there are three constraints:

- Time: 2min 30s between contact component A and component B
- Maximal difference between 2 components is 0.1g.
- Measuring weight during dispensing is inaccurate and is thus avoided.
- Dispensing is irreversible

Furthermore, weighing is a lengthy process ($\approx 13.6s$). To achieve the time requirement, there should be as little measurements as possible.

If the pump were to try to dispense the total volume in one go there would be a large deviation from the target, a lower dispensed volume would be ideal but a higher volume is detrimental. To limit this initial overshoot, the pump is set to dispense 95% of the required volume.

After this initial dispensing step, the dispensed volume is measured and the machine is calibrated. Now the next dispense still has the same problem as the initial dispense so every time, the dispensing is only 80% of the remaining required volume (95% resulted in a large overshoot).

When the load cell measures a weight $> \text{target} - 0.05\text{g}$, the dispensing will stop. If there is another component to be dispensed this process is repeated, otherwise the user will be able to take the output of the machine.

The feedforward dispensing is dependent on many uncontrollable factors:

- Time between last movement of the silicone
- Temperature
- Tube life time
- Pump specific tolerances
- Amount of gas trapped in the silicone
- Container fill rate

Therefore it is necessary to **calibrate the dispensing frequently**, this is why the first step is used for calibration. After a long time of non use, manual calibration should still be done.

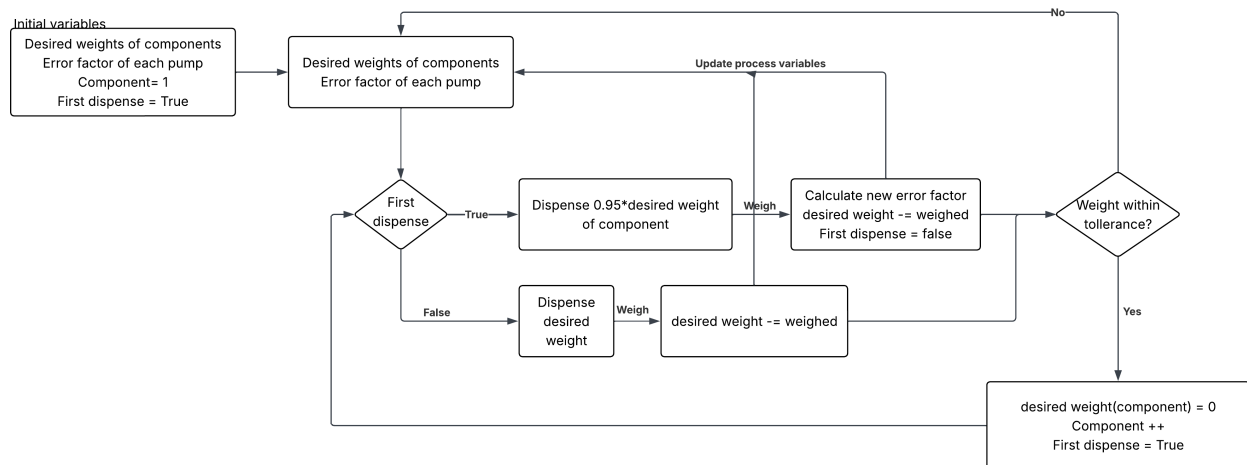


Figure 41: Dispensing Procedure Flowchart

3.7.3 Weighing

For weighing an AR0.2[23] load cell from bienfait is used, see Figure 42. This is a Load cell with high linearity (0.03%), a 1mv/V sensitivity and a maximal capacity of 200g and 12V. This mV output is amplified with a HX711[24] amplifier and a gain of 128dB. This output is converted to a digital signal with amplitude 3.3V and read out by the RPI4 [22]. The load cell is implemented in a module with a platform and mounted to the framing, see Figure 43

As the data is expected to be noisy. Thirty consecutive measurements are taken to mitigate all Gaussian noise and still obtain an adequate result. The silicone also influences the values obtained from the load cell when falling in the bucket, therefore there is a delay of 5 seconds before accepting weight values.

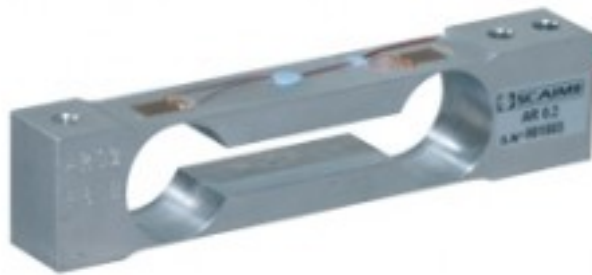


Figure 42: Type AR 0.2 Cantilever Load Cell [23]

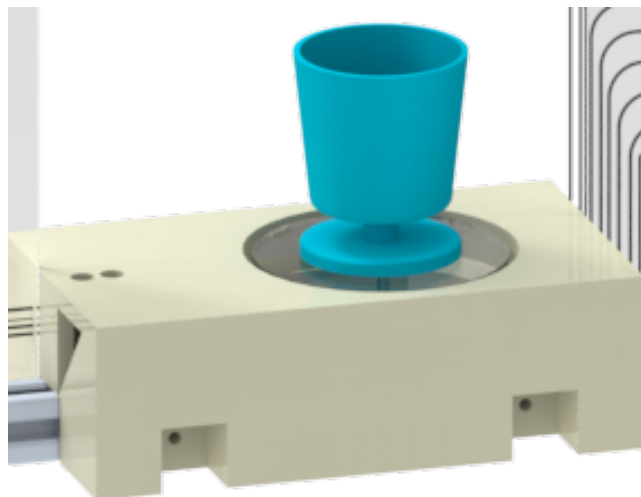


Figure 43: Load Cell Integrated in Module Mounted to the Frame

4 Conclusion

Here the final conclusions about the specifications are made. In section 4.1 the performed tests are explained. Section 4.2 contains the interpretation of the results and yet to perform analysis. Lastly the cost is evaluated in Section 4.5.

4.1 Tests

This Subsection concludes with some tests which determine the capability and precision of the machine. First, the test setup is discussed, afterwards the data and relevant values are included.

4.1.1 Test setup

To evaluate the functionality of the machine some tests were done. A target weight was selected and after the machine dispensed it, the result was evaluated using the load cell data and a laboratory scale with precision = 0.01g. Shore 5 Silicone [26] was used for these tests to limit the strain on the machine.

For the data analysis scatter plots, mean values and standard deviations were used. The most important quantitative indicators are the process capability indices: Cp and Cpk values. They are calculated using the user specification (Upper and lower specification limit (USL and LSL)).

- Cp checks if the process would be capable with perfect calibration (the capability of the process with $\mu = 0$).
 - $Cp = (USL - LSL_{Limit}) / (6 * \sigma)$
 - $Cp < 1 \Rightarrow$ not capable
 - $Cp = 1 \Rightarrow$ barely capable; 99.73% of output is correct
 - $Cp > 1 \Rightarrow$ capable;
 - $Cp \geq 1.33 \Rightarrow$ industry norm; 99.99% of output is correct
- Cpk is the actual process capability and checks the amount of outputs that are between the limits.
 - $Cpk = \min[(USL - \mu) / (3 * \sigma), (\mu - LSL) / (3 * \sigma)]$
 - $Cpk < 1 \Rightarrow$ not capable
 - $Cpk = 1 \Rightarrow$ barely capable; 99.73% of output is correct
 - $Cpk > 1 \Rightarrow$ capable;
 - $Cpk \geq 1.33 \Rightarrow$ industry norm; 99.99% of output is correct

The lower and upper specifications used are -0.1g and 0.1g.

4.1.2 Test results

Load cell accuracy and precision:

Figure 44 plots the difference between the output of the lab scale and the load cell. This can be used to evaluate the load cell precision and accuracy.

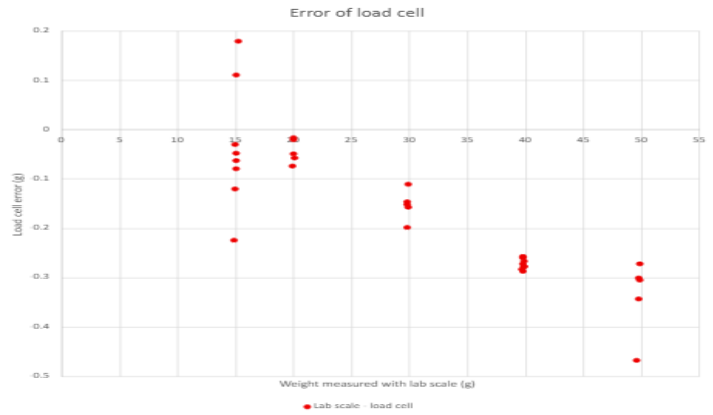


Figure 44: Scatter Plot of Measured Error of Load Cell

The x-axis of Figure 44 shows the weight measured by the lab scale and the y-axis shows the deviation of the load cell. The mean is $\approx -0.17g$ and the standard deviation is $\approx 0.14g$.

Control accuracy and precision:

To evaluate the accuracy and precision of the controlled pump, the weight measured by the load cell is plotted. This load cell output is the feedback the control uses and can be used to evaluate the pump and control precision independently of the load cell capabilities. This data can be seen in Figure 45.

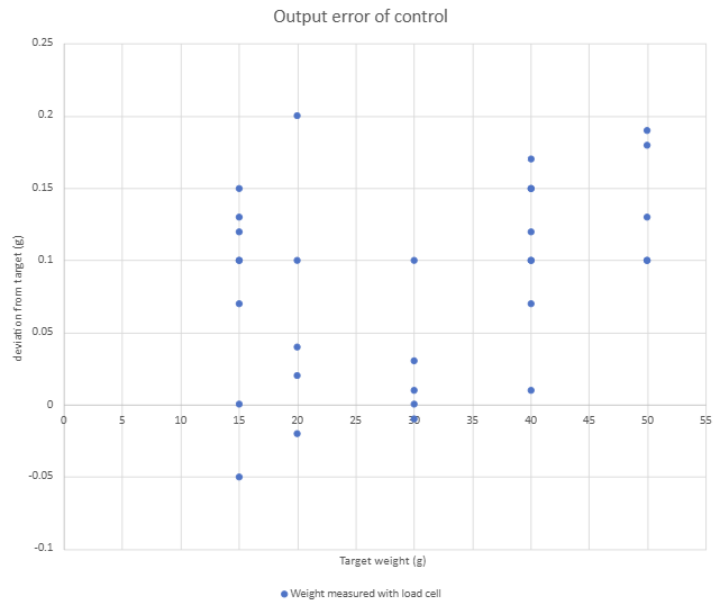


Figure 45: Scatter Plot of the Dispensing Error Measured by Load Cell

The x-axis shows the target weight (the input variable for the pump), and the y-axis shows the deviation from that target. The errors are distributed in a range of $-0.05g$ to $0.2g$, of the target weight. It may look as if the lower bound is steadily increasing but no linear fit was found.

The mean is $\approx 0.086g$ and the standard deviation is $\approx 0.067g$.

$C_p \approx 0.50$ and $C_{pk} \approx 0.070$.

Complete system accuracy and precision:

To evaluate the total accuracy and precision of the system the target weight with reference to the weight of the laboratory scale is plotted. This data is displayed in Figure 46

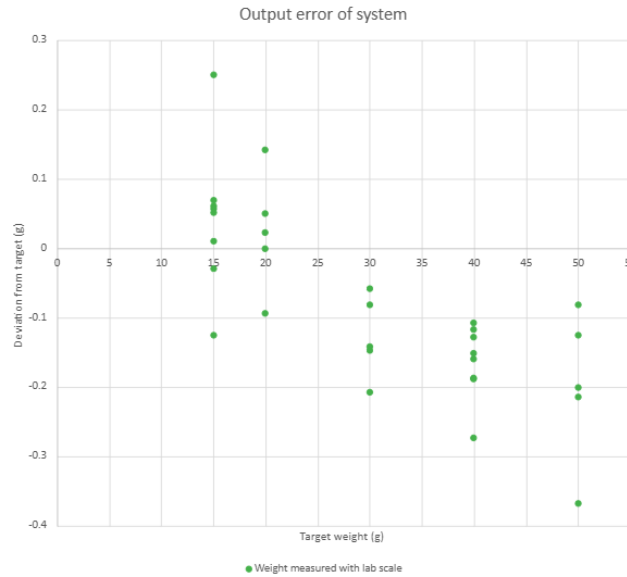


Figure 46: Scatter Plot of the Dispensing Error Measured by Laboratory Scale

Again, the x-axis shows the desired target and on the y axis the deviation from this target is visible. There is a downward trend in errors, at higher dispensing volumes the dispensing amount was too little but at lower targets it was mostly too much. The mean is $\approx -0.080g$ and the standard deviation is $\approx 0.129g$. $C_p \approx 0.257$ and $C_{pk} \approx 0.052$.

4.2 Discussion

4.2.1 Load cell

The mean difference between the load cell and the weighing scale being more than 0.1g, points to a calibration error. After calibration, the mean should be zero. This calibration error could be due to time dependent effects, such as creep. The required recalibration frequency for precise measurements is yet to be tested.

The steady downward trend of the error may be due to an error in linearity of the load cell, this effect could be decreased by taking multiple calibration points and working multiple linear segments that are calibrated separately. This is a large problem for the project as the deviation from the mean is a more important factor than the mean for silicone dispensing.

4.2.2 Control

The mean not being 0g but 0.08g is expected. This behavior is due to the pump dispensing until the measured weight $> 0.05g$. If the pump resolution were 0.1g the mean would be zero.

But even if this were the case, the large standard deviation (0.0671g) would still lead to an inaccurate machine ($C_p \approx 0.50$).

The range of results being approximately 0.25g, points to this being the resolution of the pump. This resolution could be improved by a more advanced control system influencing the silicone to make smaller droplets. Or by having an even smaller inner diameter of the output tube.

4.3 Total System

In Figure 46, it is clear that the load cell inaccuracy directly adds on to the total error leading to even larger errors. The control and resolution of this machine are not sufficient for the specifications, the

largest contributor to errors is the load cell. Making a better scale could reduce the total error variation to 0.25g instead of 0.618g.

4.4 Future Works

Test:

The variation of the load cell in time should be tested (weighing every day) to obtain a calibration frequency. More measurements for one specific weight should be done to be able to get the standard deviation at this measurement and the linearity of the load cell.

Test should be done with two pumps to validate the desired target, and to note the differences between pumps and how they affect the error. Tubing lifetime analysis should be done: measure error resulting from tube fatigue and note catastrophic tube failure time.

Design Changes:

When dispensing, the pressure build-up happens faster than the silicon can lower it with movement. Long continuous dispenses would therefore lead to problems such as the swelling seen in Figure 47. Modelling of this pressure gradient could lead to a better control algorithm and a maximal speed threshold for the stepper rotation.

This swelling is detrimental for the tube lifetime so other pressure mitigating strategies should be considered to be used such as a hydraulic accumulator, to reduce the swelling of the tube.

As seen in Figure 47, a low stiffness of the tube leads to undesired swelling. This resulted in all the tests needing to be done with shore 5 silicone. In the future when new and stiffer tubing arrives more viscous silicones should also be tested.

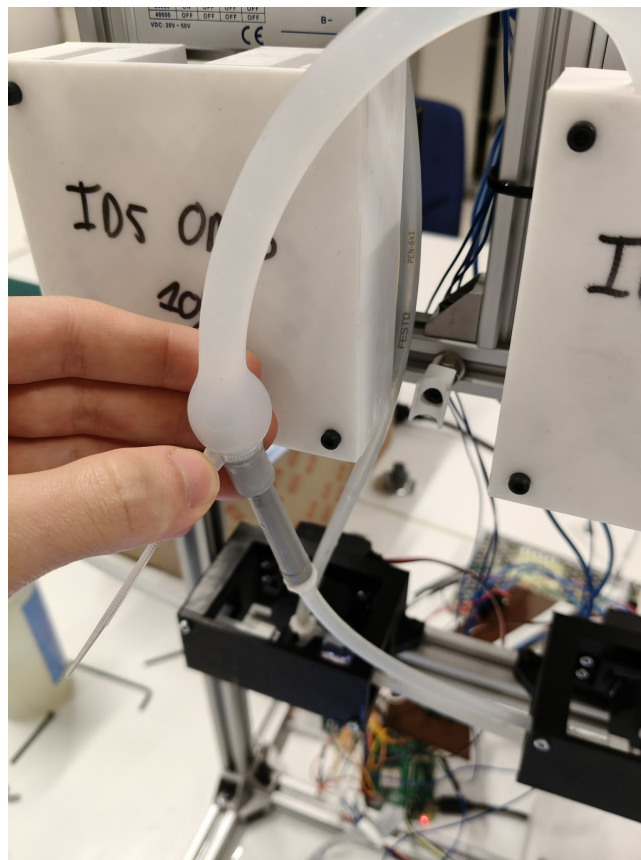


Figure 47: Tube Swelling due to too Large Pressure Buildup Necessary for Pumping 50A Silicone at a Certain Flowrate (too viscous)

4.5 Cost

The total list of the parts and the cost is listed in Table 6. With a total of 855.70 euro, the design is within cost limitations.

Table 5: Bill of Materials (BOM)

Component	Amount	Price per unit	Total price
Nema 23 stepper en driver	4	29.97	119.88
Stappenmotor beugel	4	2.05	8.20
Kogellager 6082Z	32	0.60	19.20
PLA Pumping Unit	0.45	22.50	10.125
PLA Containers	0.4	22.50	9.00
PLA Circulation Unit	0.2	22.50	4.50
Stepper motor and controller	4	10.00	40.00
Stopcock Valve	4	2.65	10.60
Screen	1	49.51	49.51
Raspberry PI 4	1	56.56	56.56
Load cell	1	215.00	215.00
Servo	4	27.20	108.80
Capacitor	2	7.20	14.40
Encoder	1	2.75	2.75
Buck converter	2	6.66	13.32
Perf board	3	4.79	14.37
Screw terminal	2	0.75	1.50
Power supply	1	37.50	37.50
HX711	1	11.16	11.16
LDO	1	10.95	10.95
Silicone ubing 5mmID 10mmOD 2m	1	34.48	34.48
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zil-verkleurig geanodiseerd/ 600mm/ geen schroefdraad	4	3.29	13.16
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zil-verkleurig geanodiseerd/ 730mm/ geen schroefdraad	2	4.01	8.02
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zil-verkleurig geanodiseerd/ 300mm/ geen schroefdraad	3	1.65	4.95
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zil-verkleurig geanodiseerd/ 250mm/ geen schroefdraad	4	1.37	5.48
Verbindingshoek 2020	25	0.29	7.25
Indraaimoer. T-sleuf 8 M4	85	0.16	13.60
Afdekkap 2020	6	0.24	1.44
Plexi (PMMA) Plate 600 x 240 x 2 mm	2	5.00	10.00
TOTAL			855.705

5 Disclaimer

Artificial intelligence tools were used in the preparation of this report to assist with improving the clarity, structure, and language of the written content. These tools served solely as a support mechanism and did not contribute to the ideas, analyses, interpretations, or conclusions presented, which represent independent academic work. Critical thinking was not replaced by the use of AI, and all information, data, and references were independently verified and cited in accordance with academic standards.

A Framework for Pump Housing in Matlab

```
1 clear all
2 close all
3 clc
4 %% Define dimensions rotor/bearings... all in mm
5 ID_bearing = 8;
6 OD_bearing = 22;
7 OD_Rotor = 56;
8 OD_Pegsupport = 10;
9 %%%TUBE PARAMETERS - TO BE CHANGED%%%
10 OD_Tube = 12; %outer diameter of the tube
11 ID_Tube = 6; %inner diameter of the tube
12 Occlusion = 0.1; %occlusion factor as a percentage
13 Rest_Tube = (OD_Tube - ID_Tube)*(1- Occlusion)
14 %% Dimensions pump housing
15 Diameter_Housing_Clearance = OD_Rotor +2 %max 1mm clearance between housing
    and rotor because m4 screws need to sit at fixed distance
16 Total_Radius_Rotor = OD_Rotor/2 - OD_Pegsupport/2 + OD_bearing/2 %from
    center of rotor to end of bearing surface
17 Diameter_OuterRing_Top = Diameter_Housing_Surface-1 %1mm clearance for top
    plate to ensure good fit+To keep tube in place (not creeping up)
18 Diameter_InnerRing_Top = OD_Rotor +4 %2mm clearance between inner ring of
    top plate and rotor, good fit/overlap with open bearing surface to
    ensure tube stays in place; but not too tight to have rubbing
19
20 %%%RESULTING DIMENSIONS TO CHANGE IN CAD%%%
21 Diameter_Housing_Surface = Total_Radius_Rotor*2 + Rest_Tube*2 %Change This
    value in the CAD design to accomodate for different occlusions and tubes
22 Pinch_Groove = OD_Tube - 3; %Change this width in the CAD design to
    accomodate for different tubes
```

Hose Pipes & Flexible Tubing



Features

- Non-yellowing
- High-tear resistance
- Suitable for most food applications
- WRAS and FDA approved

RS PRO Platinum Cured Silicone Tubing



RS PRO Professionally Approved Products bring to you professional quality parts across all product categories. Our product range has been tested by engineers and provides a comparable quality to the leading brands without paying a premium price.

Product Description

RS PRO Silicone tubing in platinum cured translucent is designed for a variety of pump and transfer applications including pharmaceutical, laboratory, bioprocess manufacturing and food and beverage applications.

- *The tubes are able to withstand a high temperature of 200°C (intermittent)*
- *Available in various internal diameters and wall thickness*

General Specifications

Material	Silicone
Food Grade	Yes
Colour	Translucent
Peristaltic Pump Compatible	Yes
Application	Laboratory, Bioprocess Manufacturing, Food and Beverage, Peristaltic Pump and Transfer Applications

Mechanical Specifications

Properties	Value	Test Method
Density	1.15 g/cm ³	
Shore A Hardness	60 ShA	D53505
Tensile Strength At Break	10.3 N/mm ²	DIN 53504S1
Elongation At Break	600%	DIN 53504S1
Tear Strength	25 N/mm ²	ASTM D624 B
Impact Resistance	60%	DIN 53512
Compression Set	20%	DIN ISO 815-B [22h @ 175°C]
Specific Gravity	1.15 g/cm ³	DIN EN ISO 1183-1A

Operation Environment Specifications

Minimum Operating Temperature	-40 °C
Maximum Operating Temperature	200 °C

Approvals

Compliance/Certifications	FDA, BfR compliant, full WRAS approval for hot and cold water and 1935/2004 compliant material used.
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Available Sizes

Stock No.	Internal Diameter (mm)	Wall Thickness (mm)	Outside Diameter (mm)	Length (m)
2732487	0.5	0.5	1.5	10
2732488	0.5	0.8	2.1	10
2732489	0.8	0.8	2.4	10
2732509	1	0.5	2	10
2732510	1	1	3	10
2732490	1.5	0.5	2.5	10
2732491	1.5	0.75	3	5
2732492	1.5	1	3.5	10
2732493	1.6	0.8	3.2	10
2732494	1.6	1.6	4.8	5
2732515	2	1	4	10
2732516	2	2	6	5
2732511	2.4	0.8	4	5
2732521	3	1	5	5
2732520	3	1.5	6	5
2732522	3	2	7	2
2732517	3.2	0.8	4.8	5
2732519	3.2	1	5.2	5
2732518	3.2	1.6	6.4	5
2732525	4	0.8	5.6	5
2732527	4	1	6	5
2732526	4	1.5	7	5
2732528	4	2	8	2
2732529	4	3	10	2
2732523	4.8	0.8	6.4	5
2732524	4.8	1.6	8	5
2732531	5	1	7	5
2732530	5	1.5	8	5
2732533	5	2	9	2

Hose Pipes & Flexible Tubing



2732532	5	2.5	10	2
2732539	6	1	8	2
2732538	6	1.5	9	2
2732540	6	2	10	2
2732541	6	3	12	2
2732534	6.3	1.6	9.5	2
2732535	6.3	2.4	11.1	2
2732536	6.3	3.2	12.7	1
2732537	6.4	0.8	8	2
2732542	7	1.5	10	2
2732543	7	2	11	2
2732544	8	1.6	11.2	2
2732547	8	2	12	2
2732545	8	2.4	12.8	2
2732546	8	2.5	13	2
2732548	8	3	14	1
2732549	8	4	16	1
2732553	9	2	13	2
2732554	9	3	15	1
2732550	9.5	2	15.9	1
2732551	9.6	1.6	12.8	2
2732552	9.6	2.4	14.4	2
2732495	10	1	12	2
2732497	10	2	14	2
2732496	10	2.5	15	1
2732498	10	3	16	1
2732499	10	4	18	1
2732503	12	2.4	16.8	2
2732500	12.7	3.2	19.1	1
2732501	12.8	1.6	16	2
2732502	12.8	2.4	17.6	1
2732504	13	2	17	1
2732505	15	3	21	1
2732506	16	2.4	20.8	1
2732507	16	3	22	1
2732508	19	3.2	25.4	1
2732512	20	3	26	1
2732513	22	3	28	1
2732514	25	3	31	1

C Datasheet Tygon Chemical Tubing [28]

TYGON® and VERSILON™ chemical hoses



TYGON® CHEMICAL

Chemical and peristaltic pump hose Material:

polyolefin / TPE (Norprene®

thermoplastic elastomer)

Operating temp: -60°C to + 74°C

Brittleness temperature: - 60°C

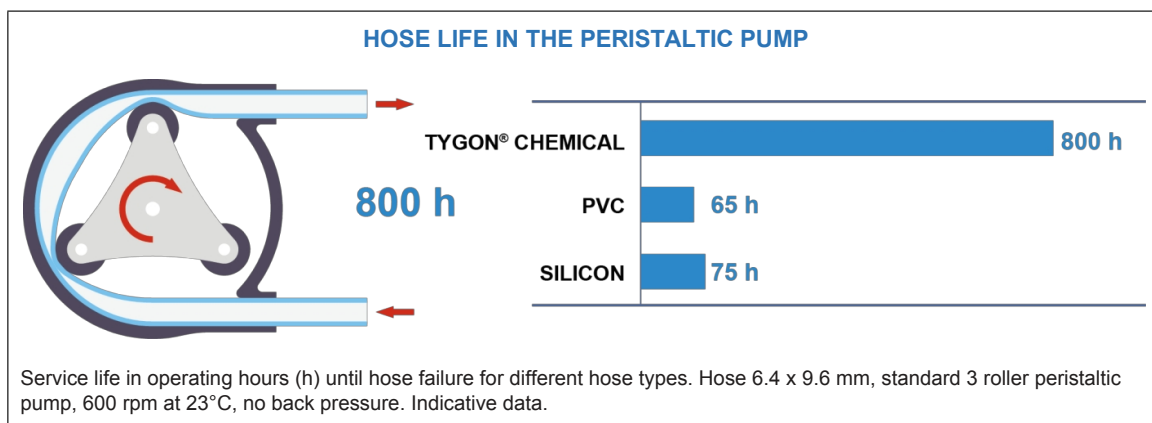
Hardness: 61° Shore (A)

Density: 0.98 g/cm³

A creamy, flexible chemical hose with a multi-layer construction specifically designed to combine chemical resistance with long life in a peristaltic pump. Smooth inner layer with very good chemical resistance, plasticiser-free, resistant to acids, bases, salts, alcohols and ketones, with low absorption and adsorption of water-based liquids. The outer layer of thermoplastic elastomer (Norprene®) provides excellent properties and long service life in peristaltic pump operation. The hose meets hygiene and food contact requirements: US FDA 21 CFR 177.1520 and is resistant to most cleaning agents and disinfectants. Sterilisation by steam, ethylene oxide or radiation. Ideal for dispensing soaps and detergents, laundry and dishwashing detergents. Used for the production and dispensing of a wide variety of chemicals in all areas of industry. Full hose reel length 15m.

index	internal diameter [mm]	external diameter [mm]	wall thickness [mm]	operating pressure* 23°C / 71°C [bar]	vacuum 23°C / 71°C [mm Hg]	bend radius [mm]
VE-AD300007	3,2	6,4	1,6	2,28 / 0,97	760 / 760	12,7
VE-AD300012	4,8	8	1,6	2 / 0,69	760 / 760	25,4
VE-AD300017	6,4	9,6	1,6	1,38 / 0,55	760 / 381	31,8
VE-AD300027	9,6	12,7	1,6	0,97 / 0,34	635 / 127	50,8
VE-AD300038	12,7	19,1	3,2	1,1 / 0,55	760 / 635	57,2

* - working pressure determined in relation to burst pressure with a safety factor of 1:5



D Datasheet Servomotor[20]



FEETECH RC Model Co.,Ltd.

产品规格书

Specification of Product

V1.0 Page 1/2

产品名称: 6V 6公斤模拟连续转舵机 Product Name: 6V 6kg.cm Analog Continuous rotation Servo
产品型号 Model No.FS5106R

1. 使用环境条件 Apply Environmental Condition

No.	Item	Specification
1-1	存储温度 Storage Temperature Range	-30℃~80℃
1-2	运行温度 Operating Temperature Range	-15℃~70℃

2. 测试环境 Standard Test Environment

No.	Item	Specification
2-1	温度 Temperature range	25℃ ±5℃
2-2	湿度 Humidity range	65%±10%

3. 机械特性 Mechanical Specification

No.	Item	Specification
3-1	尺寸 Size	A: 40.8mm B: 20.1mm C: 38mm D: 49.5mm
3-2	重量 Weight	38.6g
3-3	齿轮类型 Gear type	Plastic Gear
3-6	机构极限角度 Limit angle	360°
3-7	轴承 Bearing	2 Ball bearings
3-8	出力轴 Horn gear spline	25T
3-9	摆臂 Horn type	Plastic, POM
3-10	外壳 Case	Nylon & Fiberglass
3-11	舵机线 Connector wire	300mm ±5 mm
3-12	马达 Motor	Metal brush motor
3-13	防水性能 Splash water resistance	NO

4. 电气特性 Electrical Specification (Function of the Performance)

No.	工作电压 Operating Voltage Range	4.8V	6V
4-1*	静态电流 Idle current (at stopped)	5mA	7mA
4-2*	空载速度 No load speed	55RPM	60RPM
4-3*	空载电流 Running current (at no load)	160 mA	190 mA
4-4	堵转扭矩 Peak stall torque	5kg. cm	6kg. cm
		69.56oz. in	83.47oz. in
4-5	堵转电流 Stall current	980 mA	1100mA

Note: "*"definition is average value when the servo running with no load



FEETECH RC Model Co.,Ltd.

产品规格书

Specification of Product

V1.0 Page 2/2

产品名称: 6V 6公斤模拟连续转舵机

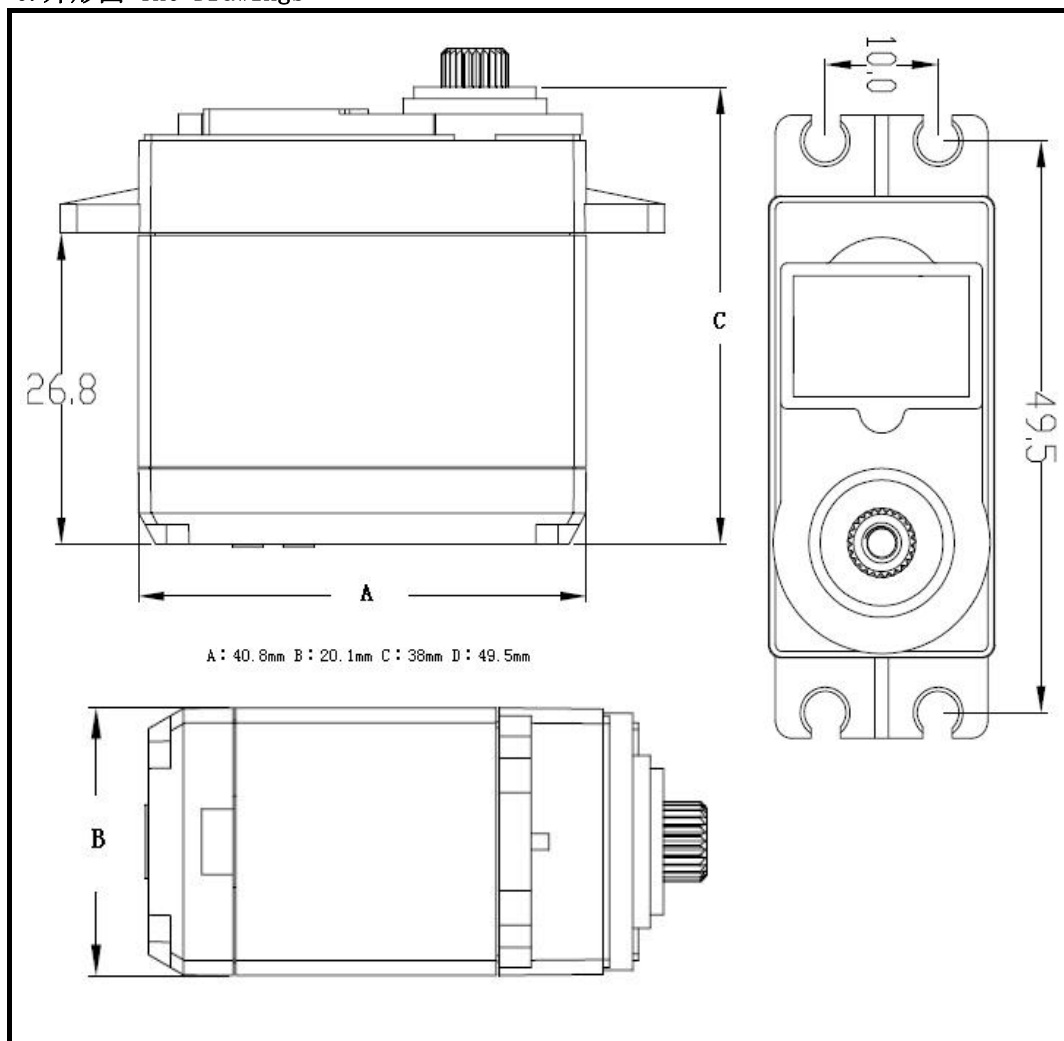
Product Name: 6V 6kg.cm Analog Continuous rotation Servo

产品型号 Model No.FS5106R

5. 控制特性 Control Specification:

No.	Item	Specification
5-1	控制信号 Command signal	Pulse width modification
5-2	放大器类型 Amplifier type	Analog comparator
5-3	脉冲宽度范围 Pulse width range	700~2300μsec
5-4	停止位置 Stop position	1500 (±5) μsec
5-5	旋转角度 Running degree	Continuous rotation
5-6	死区宽度 Dead band width	10 μsec
5-7	旋转方向 Rotating direction	CW(when 1500~700 μsec) CCW(when 1500~2300 μsec)

6. 外形图 The Drawings



E Bill of materials (BOM)

Table 6: Bill of Materials (BOM)

Component	Amount	Price per unit	Total price
Nema 23 stepper en driver	4	29.97	119.88
Stappenmotor beugel	4	2.05	8.20
Kogellager 6082Z	32	0.60	19.20
PLA Pumping Unit	0.45	22.50	10.125
PLA Containers	0.4	22.50	9.00
PLA Circulation Unit	0.2	22.50	4.50
Stepper motor and controller	4	10.00	40.00
Stopcock Valve	4	2.65	10.60
Screen	1	49.51	49.51
Raspberry PI 4	1	56.56	56.56
Load cell	1	215.00	215.00
Servo	4	27.20	108.80
Capacitor	2	7.20	14.40
Encoder	1	2.75	2.75
Buck converter	2	6.66	13.32
Perf board	3	4.79	14.37
Screw terminal	2	0.75	1.50
Power supply	1	37.50	37.50
HX711	1	11.16	11.16
LDO	1	10.95	10.95
Silicone ubing 5mmID 10mmOD 2m	1	34.48	34.48
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zilverkleurig geanodiseerd/ 600mm/ geen schroefdraad	4	3.29	13.16
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zilverkleurig geanodiseerd/ 730mm/ geen schroefdraad	2	4.01	8.02
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zilverkleurig geanodiseerd/ 300mm/ geen schroefdraad	3	1.65	4.95
Alux aluminium V-slot profiel 2020 (6 mm sleuf) – Zilverkleurig geanodiseerd/ 250mm/ geen schroefdraad	4	1.37	5.48
Verbindingshoek 2020	25	0.29	7.25
Indraaimoer. T-sleuf 8 M4	85	0.16	13.60
Afdekkap 2020	6	0.24	1.44
Plexi (PMMA) Plate 600 x 240 x 2 mm	2	5.00	10.00
TOTAL			855.705

F Test Results

Table 7: Test Total System

Target Weight (in g)	load cell weight displayed (in g)	lab scale weight displayed (in g)
15	15,07	15,25
15	14,95	15,061
15	15,13	15,01
15	15,1	14,875
15	15,15	15,07
15	15,1	15,052
15	15,12	15,057
15	15	14,97
20	19,98	19,906
20	20,02	19,999
20	20,2	20,142
20	20,04	20,023
20	20,1	20,05
30	30,03	29,919
30	30,1	29,942
30	30,01	29,858
30	30	29,853
30	29,99	29,792
40	40,17	39,892
40	40,12	39,848
40	40,1	39,813
40	40,01	39,726
40	40,15	39,872
40	40,1	39,84
40	40,15	39,883
40	40,07	39,812
50	50,13	49,786
50	50,1	49,799
50	50,1	49,632
50	50,18	49,875
50	50,19	49,918

G Logbook of Each Week

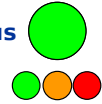
Automated Silicone Dispenser

KU LEUVEN

Planning

- **Weekly Goals:** listing all known specifications + adding additional unknowns to be filled in after meeting on Friday. Categorizing main required steps/functions to brainstorm possible solutions to make a functional decomposition.

Status



Completed Actions - Milestones

- **Completed Tasks:** Full specification list + updated with feedback; Conceptualized multiple ideas, rough outline of full functional decomposition.
- **Design Changes: /no full design yet**
- **Challenges Faced:** additional unknown info required in some cases
- **Problem-Solving:** info gathered in meeting on Friday

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Add additional ideas to functional decomposition, begin composing full concepts and evaluating them based on a value analysis; Simple graphic representations for ideas.

<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 15/02/2026

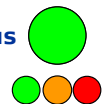
Automated Silicone Dispenser

KU LEUVEN

Planning

- **Weekly Goals:** Generate additional ideas (and graphics); Composing full concepts based on functional decomposition; Make preliminary value analysis for concepts to go over in meeting Friday.

Status



Completed Actions - Milestones

- **Completed Tasks:** Additional ideas + preliminary value analysis made; Improved preliminary value analysis based on remarks from meeting Friday with more clear categories and definitions; Last questions about exact/most important features cleared up;
- **Design Changes; /no full design yet.**
- **Challenges Faced:** lack of objective reference/comparison for value analysis.
- **Problem-Solving:** Questionnaire made to better evaluate the manual and pneumatic pouring methods relative to new ideas.

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Further diversify different concepts to clearly indicate the difference; Independently evaluate concepts, assemble scores (as averages) and compare to references from questionnaire; Meeting next Friday to indicate possible concerns/improvements and arrive at final concept. Compulsory CAE seminar on Friday.

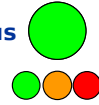
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 22/02/2026

Planning

- **Weekly Goals:** Further diversify different concepts to clearly indicate the difference; Independently evaluate concepts, assemble scores (as averages) and compare to references from questionnaire; Arrive at final concept by end of the week. Compulsory CAE seminar

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Added one additional concept (gravity); gathered results from questionnaire and assembled into scores for value analysis; Completed value analysis with diversified concepts by individual scoring and documented brief explanation for the scores.
- **Design Changes:** Peristaltic pump is selected concept (full design start next week).
- **Challenges Faced:** Possible feedback with weight sensor/flow sensor possible in budget?
- **Problem-Solving:** Additional info/websites from colleagues.

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Split up team in 2-2: Begin feedback, software and interface; More detailed cost analysis and basic calculations to indicate feasibility: peristaltic pump min/max flow + weight sensor accuracy (with ADC). Begin design to test

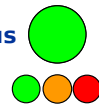
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 01/03/2026

Planning

- **Weekly Goals:** Estimated cost analysis for peristaltic pump design; Basic feasibility calculations; Begin interface and design.

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Estimated cost analysis; First peristaltic pump design; started valve design concept; Contacted bienfait about weight sensor (price, accuracy). Calculations to show feasible flow range and Torque
- **Design Changes:** Valve design changed from cam based to adaptation of manual valve.
- **Challenges Faced:** Required forces to compress flexible tubing for Torque calculation
- **Problem-Solving:** Testing of peristaltic pump design to see if Torque is adequate, else use Nema17 with higher torque.

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Framing of the design, peristaltic pump adaptations for Tolerances and fit of Bearings; Valve design; Further expanding interface. Testing further fit for Peristaltic pump housing (tubing).

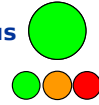
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 08/03/2026

Planning

- **Weekly Goals:** Testing tolerances/fit for Peristaltic pump housing (tubing); Expanding interface; Valve design from manual valve;

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Adapted tolerances of pump housing and rotor (for bearings); Two framing options were discussed; Tested gripping strength of flexible tubing to implement proper tolerance in pump housing
- **Design Changes:** 'Wall' framing option chosen (Clear overview).
- **Challenges Faced:** No response yet from Bienfait about weight sensor; Parts needed for testing.
- **Problem-Solving:** Looking for other suppliers for weight sensors in case of no response; Ordering parts over the Weekend;

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Adding more rollers to rotor of peristaltic pump; Testing peristaltic pump; Interface and Control of peristaltic pump; Fitting first design of Valve; Locking mechanism for tubing (luer/threaded)

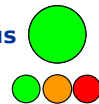
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 15/03/2026

Planning

- **Weekly Goals:** Adding more rollers to rotor of peristaltic pump; Testing peristaltic pump; Interface and Control of peristaltic pump; Fitting first design of Valve; Locking mechanism for tubing (luer/threaded); Completing safety training for access to the lab.

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Tested peristaltic pump with water (successfully); First design of silicone container; Safety training completed; Weight sensor from Bienfait added to budget (+processing with Hx711); Start of assembly design;
- **Design Changes:** Luer Lock mechanism added to a first version of the container (instead of threaded)
- **Challenges Faced:** Non-User-friendly pump assembly with sliding mechanism to change tubing; Material ordered from 123-3D not delivered
- **Problem-Solving:** change design from sliding assembly to top lid design.

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Testing new version of peristaltic pump with tubing from RS and with silicone; Further design of full system assembly; Integration of control and interface code for Raspberry pi

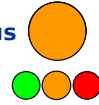
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 21/03/2026

Planning

- **Weekly Goals:** Testing new version of peristaltic pump with tubing from RS and with silicone; Further design of full system assembly; Integration of control and interface code for Raspberry pi

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Tested peristaltic pump in different configurations with silicone: harder tubing, softer tubing, multiple occlusions, different number of rollers; Also tested other commercial (bigger) peristaltic pump with same silicone but even more (different) kinds of tubing (natural big rubber flexible, very hard smaller ID). Tested valve successfully.
- **Design Changes:** Probably change to bigger motor; Valve dimensions and tolerances finalized.
- **Challenges Faced:** Viscous silicone causes tubing to stay flattened and almost zero pumping with large pulsating flow; Torque of motor seems too little for proper pumping sometimes.
- **Problem-Solving:** Contacted multiple (4) companies to inform about different tubing sizes/flexibility for further testing

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Prepare presentation for end of next week Friday with overview of work done until now. Testing further with gravity assist from container. Compose further plan of attack after presentation and ordering parts for after easter break.

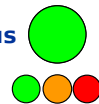
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 29/03/2026

Planning

- **Weekly Goals:** Presentation on Friday; Testing with gravity assist from container. Planning after presentation and ordering parts. Finish full assembly view.

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Positive from presentation to order parts; Further contact with Tubes International. Already ordered some more standard silicone tubes from RS components
- **Design Changes:** changed plans for peristaltic pump to use much bigger Nema 23 instead of 17 to allow for testing of various tubes without torque restrictions; idem for servos for the valves.
- **Challenges Faced:** Uncertainty for some parts as design for eg pumps can still change.
- **Problem-Solving:** As framing, electrical enclosing... are less important, these are put on low priority.

Ongoing Actions - Next steps

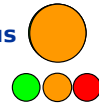
- **Plans for the Upcoming Weeks (Easter break) :** Further contact with Tubes International to obtain samples of various tubes for peristaltic pump. Change design of peristaltic pump framework to allow Nema 23 stepper integration and easy changing of parameters to test samples. When parts arrive new motor control, integrate weight sensor...

<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Piot>

Datum: 5/04/2026

Planning

- **Weekly Goals:** Assess current position and make planning for the remaining weeks. Test new tubes from RS components with new Nema23 motor peristaltic pump design after implementation of new controller with Nema 23. Start integrating weight sensor; Start report.

Status**Completed Actions - Milestones**

- **Completed Tasks:** Successfully tested peristaltic pump with 5_ID10_OD and 6_ID12_OD silicone tubes from RS; Flow rate was almost at the desired level; Motor control and design are thus almost finalized.
- **Design Changes:** First protective design for weight sensor; new design for rotor of different valves that have a bigger inner diameter to minimize pressure needed after the pump.
- **Challenges Faced:** Excessive pressure buildup in tube with 5_ID10_OD due to restriction
- **Problem-Solving:** Adapt connectors after the tube of the pump to have a smooth transition to the valve restriction and use of bigger hard tubing directly behind the pump

Ongoing Actions - Next steps

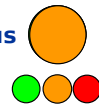
- **Plans for the Upcoming Week:** Finish valve design; finish weight module design; finish container design; Further testing of peristaltic pump with 5 instead of 4 rollers; report; weight sensor integration with HX711; Order additional parts since some designs are fixed (eg 2 more Nema 23's and controllers, more bearings; threaded inserts).

<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 26/04/2026

Planning

- **Weekly Goals:** Finish Valve design; weight module design; container design and peristaltic pump; Test pump with gradual restrictor for better flow. Integration of all code in overall system (raspberry pi) Framing redesign and parts;

Status**Completed Actions - Milestones**

- **Completed Tasks:** Tested peristaltic pump with restrictor; Redesigned framing for the different pumps, controllers, display and provided space for electronics. Added markings to the containers to give indication of remaining volume.
- **Design Changes:** Tube with 5_ID10_OD turned out to be sufficient with the gradual restrictor and additional testing showed a great influence of the suction side restrictions on the flow rate as well → redesigned container connection to be a big slip connection instead of luer lock (5mm<->3mm previous). I
- **Challenges Faced:** Rotor did not stay attached to motor with the 6mm ID tube due to too high torque required: heat set inserts are pushed out of the PLA due to excessive needed force; test with 5 rollers could not be performed
- **Problem-Solving:** Switched to 5_ID10_OD tube that turned out to be sufficient; with full setup (restrictor, valve, tubes) the fluctuating output turned out to be not too big (<-> speed) → 5 rollers probably not even necessary. Adapted the design of the rotor to include D notch for additional strength (and to be printed out of stronger filament).

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Final tests with restrictor, new container opening, adapted rotor... Assemble weight module to perform integrated tests + test code. Order parts for the full framing. Report.

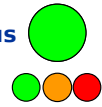
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 03/05/2026

Planning

- **Weekly Goals:** Final tests with restrictor, new container opening and adapted D lock rotor; Assembly of weight module, order full framing parts, report, test code weight sensor + integrate with the interface code and control scheme.

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Final Weight module assembled (laser cut platform + printed housing); Ordered full framing; Tested pump control with interface (open loop, no feedback from weight module yet). All valves assembled, Cover and mounting. Final required (major) parts printed.
- **Design Changes:** Slight Adaptation of weight module to allow more bending of sensor)
- **Challenges Faced:** Problems with weight sensor HX711 amplifier (bad module). Motors get to ~50°C, meaning PLA μ which is used can deform (around it's T_glass).
- **Problem-Solving:** Fixed board problems, now functional weight outputs. Switched Pump housings to PETG which has much higher T_glass ~80°C to be safe and avoid deformation of pump housing (and changing occlusion...)

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Assemble framing, add electronics, pumps, valves... to the framing to have fully assembled system (for 2 proof of concept components). Integrated tests with feedback from now working Weight sensor., which needs to be tuned first. Design casings for screen + control buttons and miscellaneous small parts to be covered.

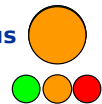
<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 10/05/2026

Planning

- **Weekly Goals:** Assemble framing and add all parts, solder electronics, integrated tests, report.

Status

**Completed Actions - Milestones**

- **Completed Tasks:** Full framing assembled + all modules added. Electronics soldered.
- **Design Changes:** /
- **Challenges Faced:** Bugs in control system.
- **Problem-Solving:** Remake some parts of code, debugging.

Ongoing Actions - Next steps

- **Plans for the Upcoming Week:** Finish Report, integrated testing, preparing presentation

<Kamiel Vandekerckhove> <Gregor Van Vlierberghe> <Niels Quackels> <David Plot>

Datum: 17/05/2026

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